

Project Data Page:





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Nozzle Calcs.: K3-2"	100
Nozzle Calcs.: K4-2"	105
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Nozzle Calcs.: K6-2"	115
Nozzle Calcs.: K7-2"	120
Nozzle Calcs.: K8-2"	125
Nozzle Calcs.: K9-2"	130
Nozzle Calcs.: K10-2"	135
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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Feed Gas Knockout Drum (174-D)Rev0- 7%.pvdb

Date of Analysis : Sep 16, 2026 2:09pm

PV Elite 28 U1, March 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1825.000 mm. ID
 Vessel Design Length, Tangent to Tangent 2450.00 mm.
 Distance of Bottom Tangent above Grade 3300.00 mm.
 Distance of Base above Grade 300.00 mm.
 Specified Datum Line Distance 3000.00 mm.
 Internal Design Temperature 82 °C
 Internal Design Pressure 63.996 bars
 External Design Temperature 0 °C
 Maximum Allowable Working Pressure 64 bars
 Shop Test Pressure 89.799 bars
 Field Test Pressure 83.195 bars
 Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature -22.0 °C
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-266 2	...	...	K03506	+IT
Nozzle	SA-105	...	...	K03504	
Nozzle Flg	SA-105	...	...	K03504	
Basing	SA-516 70	...	...	K02700	
Base Bolting	SA-307 A	...	6 ≤ t ≤ 100		

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
SKIRT	...	...	...	1.0000	N/A	No
SKIRT	...	...	...	1.0000	N/A	No
BOTTOM HEAD	64.140	0.00	64.600	3.0000	No	No
SHELL	64.091	0.00	65.800	3.0000	N/A	No
TOP HEAD	64.003	0.00	64.800	3.0000	No	No

Liquid Level: 1406.25 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 0.998

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Skirt	-1000.0	2000.0	10.0	10.0	...	...	0.70	0.70
Skirt	0.0	1000.0	10.0	10.0	...	...	0.70	0.50
Ellipse	50.0	50.0	54.0	46.0	45.6	...	1.00	1.00



Cylinder	2400.0	2350.0	48.0	48.0	46.7	...	1.00	1.00
Ellipse	2450.0	50.0	54.0	46.0	45.5	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Trunnion	WRC-107/537	0.162	n/a	Passed
Trunnion	WRC-107/537	0.162	n/a	Passed
Trunnion	WRC-107/537	0.162	n/a	Passed
Trunnion	WRC-107/537	0.162	n/a	Passed
Trunnion	WRC-107/537	0.162	n/a	Passed
Trunnion	WRC-107/537	0.162	n/a	Passed
Trunnion	WRC-107/537	0.162	n/a	Passed
Trunnion	WRC-107/537	0.162	n/a	Passed
Trunnion	WRC-107/537	0.162	n/a	Passed
Trunnion	WRC-107/537	0.161	n/a	Passed
Trunnion	WRC-107/537	0.161	n/a	Passed
Trunnion	WRC-107/537	0.160	n/a	Passed
Trunnion	WRC-107/537	0.160	n/a	Passed
Trunnion	WRC-107/537	0.159	n/a	Passed
Trunnion	WRC-107/537	0.159	n/a	Passed
Trunnion	WRC-107/537	0.158	n/a	Passed
Trunnion	WRC-107/537	0.158	n/a	Passed
Trunnion	WRC-107/537	0.157	n/a	Passed
Trunnion	WRC-107/537	0.156	n/a	Passed
Trunnion	WRC-107/537	0.155	n/a	Passed
Trunnion	WRC-107/537	0.155	n/a	Passed
Trunnion	WRC-107/537	0.154	n/a	Passed
Trunnion	WRC-107/537	0.153	n/a	Passed
Trunnion	WRC-107/537	0.152	n/a	Passed
Trunnion	WRC-107/537	0.151	n/a	Passed
Trunnion	WRC-107/537	0.150	n/a	Passed
Trunnion	WRC-107/537	0.149	n/a	Passed
Trunnion	WRC-107/537	0.148	n/a	Passed
Trunnion	WRC-107/537	0.146	n/a	Passed
Trunnion	WRC-107/537	0.145	n/a	Passed
Trunnion	WRC-107/537	0.144	n/a	Passed
Trunnion	WRC-107/537	0.143	n/a	Passed
Trunnion	WRC-107/537	0.141	n/a	Passed
Trunnion	WRC-107/537	0.140	n/a	Passed
Trunnion	WRC-107/537	0.138	n/a	Passed
Trunnion	WRC-107/537	0.137	n/a	Passed
Trunnion	WRC-107/537	0.135	n/a	Passed
Trunnion	WRC-107/537	0.134	n/a	Passed
Trunnion	WRC-107/537	0.132	n/a	Passed
Trunnion	WRC-107/537	0.131	n/a	Passed
Trunnion	WRC-107/537	0.129	n/a	Passed
Trunnion	WRC-107/537	0.127	n/a	Passed
Trunnion	WRC-107/537	0.126	n/a	Passed
Trunnion	WRC-107/537	0.124	n/a	Passed
Trunnion	WRC-107/537	0.122	n/a	Passed
Trunnion	WRC-107/537	0.120	n/a	Passed
Trunnion	WRC-107/537	0.118	n/a	Passed
Trunnion	WRC-107/537	0.117	n/a	Passed
Trunnion	WRC-107/537	0.115	n/a	Passed
Trunnion	WRC-107/537	0.113	n/a	Passed
Trunnion	WRC-107/537	0.111	n/a	Passed
Trunnion	WRC-107/537	0.109	n/a	Passed
Trunnion	WRC-107/537	0.107	n/a	Passed
Trunnion	WRC-107/537	0.105	n/a	Passed
Trunnion	WRC-107/537	0.103	n/a	Passed
Trunnion	WRC-107/537	0.104	n/a	Passed
Trunnion	WRC-107/537	0.105	n/a	Passed
Trunnion	WRC-107/537	0.106	n/a	Passed
Trunnion	WRC-107/537	0.107	n/a	Passed
Trunnion	WRC-107/537	0.108	n/a	Passed
Trunnion	WRC-107/537	0.109	n/a	Passed

Trunnion	WRC-107/537	0.110	n/a	Passed
Trunnion	WRC-107/537	0.111	n/a	Passed
Trunnion	WRC-107/537	0.112	n/a	Passed
Trunnion	WRC-107/537	0.113	n/a	Passed
Trunnion	WRC-107/537	0.114	n/a	Passed
Trunnion	WRC-107/537	0.115	n/a	Passed
Trunnion	WRC-107/537	0.116	n/a	Passed
Trunnion	WRC-107/537	0.116	n/a	Passed
Trunnion	WRC-107/537	0.117	n/a	Passed
Trunnion	WRC-107/537	0.118	n/a	Passed
Trunnion	WRC-107/537	0.119	n/a	Passed
Trunnion	WRC-107/537	0.120	n/a	Passed
Trunnion	WRC-107/537	0.121	n/a	Passed
Trunnion	WRC-107/537	0.122	n/a	Passed
Trunnion	WRC-107/537	0.123	n/a	Passed
Trunnion	WRC-107/537	0.124	n/a	Passed
Trunnion	WRC-107/537	0.125	n/a	Passed
Trunnion	WRC-107/537	0.126	n/a	Passed
Trunnion	WRC-107/537	0.127	n/a	Passed
Trunnion	WRC-107/537	0.128	n/a	Passed
Trunnion	WRC-107/537	0.129	n/a	Passed
Trunnion	WRC-107/537	0.130	n/a	Passed
Trunnion	WRC-107/537	0.131	n/a	Passed
Trunnion	WRC-107/537	0.133	n/a	Passed
Trunnion	WRC-107/537	0.134	n/a	Passed
Trunnion	WRC-107/537	0.136	n/a	Passed
Trunnion	WRC-107/537	0.138	n/a	Passed
Trunnion	WRC-107/537	0.140	n/a	Passed
Trunnion	WRC-107/537	0.142	n/a	Passed
Trunnion	WRC-107/537	0.162	n/a	Passed
D1-2"	WRC-107/537	0.890	n/a	Passed
N1-16"	WRC-297	0.964	Shell	Passed
N2-16"	WRC-107/537	0.890	n/a	Passed

Wind/Earthquake Shear, Bending:

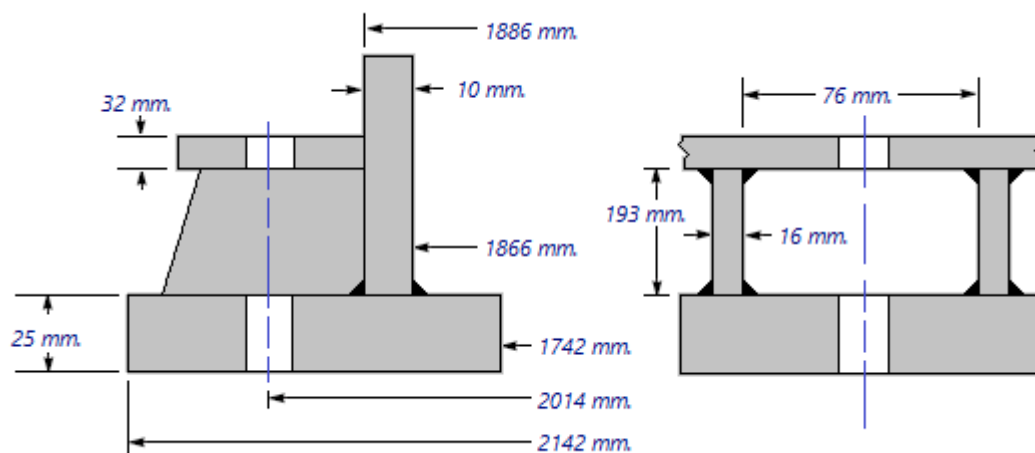
From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	1371.3	3648	5252.14	15609.1
20	30	2500	1158.47	3547.94	2722.32	8412.98
30	40	3025	1052.05	3445.54	1617.04	4916.17
40	50	4225	739.885	2817.56	1572.24	4759.59
50	60	5425	497.988	1026.57	117.711	242.654

Abs Max of the all of the Stress Ratio's : 0.4829

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	25.0000	mm.
Inside Diameter of Basing	1742.0000	mm.
Outside Diameter of Basing	2142.0000	mm.
Nominal Diameter of Bolts	36.0000	mm.
Diameter of Bolt Circle	2013.9999	mm.
Number of Bolts	12	
Thickness of Gusset Plates	16.0000	mm.
Average Width of Gusset Plates	118.0000	mm.
Height of Gussets	193.0000	mm.
Distance between Gussets	76.0000	mm.
Thickness of Top Plate or Ring	32.0000	mm.
Radial Width of the Top Plate	128.0000	mm.
Number of Gussets	24	

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	1371. Kg
Total Earthquake Shear on Support	3648. Kg
Wind Moment on Support	5300. Kg-m.
Earthquake Moment on Support	15657. Kg-m.
Maximum Initial Bolt Load	205. Kg

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	16719.6 kg.
Shop Test - Fabricated + water (full)	24716.4 kg.
Shipping - Fab. + rem. intls.+ shipping app.	16719.6 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	25369.8 kg.
Empty - Fab. + intls. + details + wghts.	25369.8 kg.
Operating - Empty + operating liquid (no ca)	28634.2 kg.
Field Test - Empty weight + test medium (full)	33136.6 kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
D1-2"	64	...	...	OK	8.26	OK	Passed
N1-16"	64	...	...	OK	12.86	OK	Passed
K1-2"	64	...	...	OK	8.26	OK	Passed
K2-2"	64	...	...	OK	8.26	OK	Passed
K3-2"	64	...	...	OK	8.26	OK	Passed
K4-2"	64	...	...	OK	8.26	OK	Passed
K5-2"	64	...	...	OK	8.26	OK	Passed
K6-2"	64	...	...	OK	8.26	OK	Passed
K7-2"	64	...	...	OK	8.26	OK	Passed
K8-2"	64	...	...	OK	8.26	OK	Passed
K9-2"	64	...	...	OK	8.26	OK	Passed
K10-2"	64	...	...	OK	8.26	OK	Passed
M1-24"	64	...	...	...		OK	Passed
N2-16"	64	...	...	OK	12.86	OK	Passed
K11-2"	64	...	...	OK	8.26	OK	Passed
K12-2"	64	...	...	OK	8.26	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 64 Nozzle : K12-2"
 Minimum MAWP Shells/Flanges : 64 Element : BOTTOM HEAD
 Minimum MAPnc Shells/Flanges : 69.2 Element : TOP HEAD

Computed Vessel M.A.W.P. : 64 bars

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
30	D1-2"	0.000	0.000	190.648
40	N1-16"	1505.000	90.000	824.800
40	K1-2"	1150.000	270.000	194.648
40	K2-2"	200.000	270.000	194.648
40	K3-2"	950.000	90.000	194.648
40	K4-2"	200.000	90.000	194.648
40	K5-2"	950.000	180.000	194.648
40	K6-2"	200.000	180.000	194.648
40	K7-2"	950.000	0.000	194.648
40	K8-2"	200.000	0.000	194.648
40	K9-2"	950.000	45.000	194.648
40	K10-2"	200.000	45.000	194.648
40	M1-24"	1050.000	225.000	1231.200
50	N2-16"	0.000	0.000	824.800
50	K11-2"	0.000	270.000	190.648
50	K12-2"	0.000	90.000	190.648

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
D1-2"	2.000 in	Actual	LWN	4.120	26.924	...	...	234.14	900
K1-2"	2.000 in	Actual	LWN	4.120	26.924	...	...	209.50	600
K2-2"	2.000 in	Actual	LWN	4.120	26.924	...	...	209.50	600
K3-2"	2.000 in	Actual	LWN	4.120	26.924	...	...	209.50	600
K4-2"	2.000 in	Actual	LWN	4.120	26.924	...	...	209.50	600
K5-2"	2.000 in	Actual	LWN	4.120	26.924	...	...	209.50	600
K6-2"	2.000 in	Actual	LWN	4.120	26.924	...	...	209.50	600
K7-2"	2.000 in	Actual	LWN	4.120	26.924	...	...	209.50	600
K8-2"	2.000 in	Actual	LWN	4.120	26.924	...	...	209.50	600
K9-2"	2.000 in	Actual	LWN	4.120	26.924	...	...	209.50	600
K10-2"	2.000 in	Actual	LWN	4.120	26.924	...	...	209.50	600
K11-2"	2.000 in	Actual	LWN	4.120	26.924	...	...	722.79	600
K12-2"	2.000 in	Actual	LWN	4.120	26.924	...	...	722.79	600
N1-16"	16.000 in	Actual	WNF	18.205	28.000	...	...	1898.00	600
N2-16"	16.000 in	Actual	WNF	18.205	28.000	...	...	571.79	600
M1-24"	24.000 in	Actual	LWN	26.205	28.000	...	...	510.98	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
D1-2"	SA-106 B	46.000	10.000	...	...	...
K1-2"	SA-105	48.000	10.000	...	...	...
K2-2"	SA-105	48.000	10.000	...	...	...
K3-2"	SA-105	48.000	10.000	...	...	...
K4-2"	SA-105	48.000	10.000	...	...	...
K5-2"	SA-105	48.000	10.000	...	...	...
K6-2"	SA-105	48.000	10.000	...	...	...
K7-2"	SA-105	48.000	10.000	...	...	...
K8-2"	SA-105	48.000	10.000	...	...	...
K9-2"	SA-105	48.000	10.000	...	...	...
K10-2"	SA-105	48.000	10.000	...	...	...
K11-2"	SA-105	46.000	10.000	...	...	...
K12-2"	SA-105	46.000	10.000	...	...	...
N1-16"	SA-266 2	48.000	10.000	...	...	...
N2-16"	SA-266 2	46.000	10.000	...	...	...
M1-24"	SA-266 2	48.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
D1-2"	...	0.0	187.32	0.00	BOTTOM HEAD
K1-2"	1150.000	270.0	160.00	0.00	SHELL
K2-2"	200.000	270.0	160.00	0.00	SHELL
K3-2"	950.000	90.0	160.00	0.00	SHELL
K4-2"	200.000	90.0	160.00	0.00	SHELL



K5-2"	950.000	180.0	160.00	0.00	SHELL
K6-2"	200.000	180.0	160.00	0.00	SHELL
K7-2"	950.000	0.0	160.00	0.00	SHELL
K8-2"	200.000	0.0	160.00	0.00	SHELL
K9-2"	950.000	45.0	160.00	0.00	SHELL
K10-2"	200.000	45.0	160.00	0.00	SHELL
K11-2"	...	270.0	650.00	0.00	TOP HEAD
K12-2"	...	90.0	650.00	0.00	TOP HEAD
N1-16"	1505.000	90.0	250.00	1600.00	SHELL
N2-16"	...	0.0	500.00	0.00	TOP HEAD
M1-24"	1050.000	225.0	350.00	0.00	SHELL



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 54.0mm. THK X 1825.0mm. ID X 50.0mm.	SA-516 70
1	INSULATION: 2000mm X 50mm THK	
1	INSULATION: 1000mm X 50mm THK	
1	PLATFORM: 1000mm X 1500mm WIDE	
2	LADDER: 3000.0mm	...
2	TRUNNIONS:	SA-106 B
2	TRUNNION PADS: 373.0mm OD X 273.0mm ID X 22.0mm THK	
1	PLATFORM: 1000mm X 4000mm WIDE	
1	CLASS 600 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 2142.0mm. OD X 1742.0 ID X 25.0mm.	SA-516 70
12	BASERING BOLTS: 36.0mm. DIA	SA-307 A
12	BASERING BOLT NUTS: 36.0mm. DIA	...
1	BASERING TOP RING: 2142.0mm. OD X 1886.0 ID X 32.0mm.	...
24	BASERING GUSSET PLATES: 118.0mm. X 118.0mm. X 193.0mm. X 16.0mm.	...
1	CYLINDRICAL SKIRT SEGMENT 10.0mm. THK X 3048.0mm. X 2000.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 10.0mm. THK X 2814.2mm. X 2000.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 10.0mm. THK X 3110.2mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 10.0mm. THK X 2751.9mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 48.0mm. THK X 2277.4mm. X 2350.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 48.0mm. THK X 2261.5mm. X 2350.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 48.0mm. THK X 1194.4mm. X 2350.0mm.	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
BOTTOM HEAD	[10]	D	-21	-22		0.988	46.000	1.00	Yes
BOTTOM HEAD	[7]	D	-18	-26		0.856	54.000	1.00	Yes
SHELL	[8]	D	-21	-22		0.971	48.000	1.00	Yes
TOP HEAD	[10]	D	-21	-22		0.988	46.000	1.00	Yes
TOP HEAD	[7]	D	-18	-26		0.856	54.000	1.00	Yes
D1-2"	[1]	B	1	-104		0.066	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-48					
N1-16"	[1]	D	-21	-22		0.973	48.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-38					
K1-2"	[1]	A	21	-104		0.056	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-38					
K2-2"	[1]	A	21	-104		0.056	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-38					
K3-2"	[1]	A	21	-104		0.056	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-38					
K4-2"	[1]	A	21	-104		0.056	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-38					
K5-2"	[1]	A	21	-104		0.056	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-38					
K6-2"	[1]	A	21	-104		0.056	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-38					
K7-2"	[1]	A	21	-104		0.056	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-38					
K8-2"	[1]	A	21	-104		0.056	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-38					
K9-2"	[1]	A	21	-104		0.056	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-38					
K10-2"	[1]	A	21	-104		0.056	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-38					
M1-24"	[1]	D	-21	-22		0.973	48.000	1.00	No
Nozzle Flg	[5]	B	-18	-38					
N2-16"	[1]	D	-21	-27		0.892	46.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-38					
K11-2"	[1]	A	21	-104		0.056	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-38					
K12-2"	[1]	A	21	-104		0.056	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-38					
Nozl. Bolting	[22]		-48			1.000	64.000	1.00	No

Warmest MDMT: 21 -22

Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature -22.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations



[21] - Body Flange Bolting Material
[22] - Nozzle Flange Bolting Material
[23] - Stiffening Ring to Shell Weld
[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note 20 40 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (barsUniCode):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	63.996	bars
Design Internal Temperature	82.0	°C
Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16		IP+VE+EW	
Load Case 17		NP+VO+OW	
Load Case 18		FS+BS+IP+OW	
Load Case 19		FS+BS+EP+OW	
Load Case 20		BL+IP+OW	
Wind Design Code		ASCE/SEI 7-16	
Wind Load Reduction Scale Factor		1.100	
Basic Wind Speed	[V]	97.2	Km/hr
Surface Roughness Category		C: Open Terrain	
Type of Surface		Moderately Smooth	
Base Elevation		300	mm.
Percent Wind for Hydrotest		50.0	
Using User defined Wind Press. Vs Elev.		N	
Height of Hill or Escarpment	H or Hh	0	mm.
Distance Upwind of Crest	Lh	0	mm.
Distance from Crest to the Vessel	x	0	mm.
Type of Terrain (Hill, Escarpment)		Flat	
Damping Factor (Beta) for Wind (Ope)		0.0100	
Damping Factor (Beta) for Wind (Empty)		0.0000	
Damping Factor (Beta) for Wind (Filled)		0.0000	
Site Elevation above Sea Level		0	mm.
Seismic Design Code		ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor		1.000	
Importance Factor		1.250	
Table Value Fa		1.300	
Table Value Fv		1.500	
Max. Mapped Res. Acceleration	[Ss]	0.147	
Max. Eff. Ground Acceleration	[S]	0.065	
Force Modification Factor R		1.250	
Site Class		C	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	2.500	
Force Factor		0.000	
Consider Vertical Acceleration		No	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Moment Reduction Factor Tau		1.000	
Overall M.A.W.P. + Static Head		Y	
Consider MAP New and Cold in Noz. Design		N	
Consider External Loads for Nozzle Des.		Y	
Use ASME VIII-1 Appendix 1-9		N	
Perform Blast Load Analysis		No	
Material Database Year	Current w/Addenda or Code Year		

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/5
Element From Node	10

Element To Node	20	
Element Type	Skirt Support	
Description	SKIRT	
Distance "FROM" to "TO"	2000	mm.
Skirt Inside Diameter	1866	mm.
Diameter of Skirt at Base	1866	mm.
Skirt Thickness	10	mm.
Internal Corrosion Allowance	1	mm.
Nominal Thickness	10	mm.
External Corrosion Allowance	0	mm.
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.37	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm2
Allowable Stress, Operating	138	N./mm2
Allowable Stress, Hydrotest	234	N./mm2
Material Density	0.00775	kg./cm3
P Number Thickness	32.004	mm.
Yield Stress, Operating	242.6	N./mm2
UCS-66 Chart Curve Designation	D	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Insulation	
Detail ID	Fire Proofing	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	2000	mm.
Thickness of Insulation	50	mm.
Density	0.0024	kg./cm3

Element#	2/5	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	SKIRT	
Distance "FROM" to "TO"	1000	mm.
Skirt Inside Diameter	1866	mm.
Diameter of Skirt at Base	1866	mm.
Skirt Thickness	10	mm.
Internal Corrosion Allowance	1	mm.
Nominal Thickness	10	mm.
External Corrosion Allowance	0	mm.
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.37	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.5	
Weld is pre-Heated	No	
Element From Node	20	
Detail Type	Insulation	
Detail ID	Fire Proofing	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	1000	mm.
Thickness of Insulation	50	mm.
Density	0.0024	kg./cm3

Element#	3/5
Element From Node	30



Element To Node	40
Element Type	Elliptical
Description	BOTTOM HEAD
Distance "FROM" to "TO"	50 mm.
Inside Diameter	1825 mm.
Element Thickness	46 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	54 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	63.996 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.37
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	0 mm.
Platform Start Angle (degrees)	360.0
Platform End Angle (degrees)	0.0
Platform Wind Area	36000 cm2
Platform Weight	2988.2 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm2
Platform Width	1500 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	85 mm.
Platform Force Coefficient	1.2
Ladder Layout Angle	180.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	3000 mm.
Unit Weight of Ladder	0.03773 Kg/mm.
Platform Length (top head platform)	0 mm.

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	-456.25 mm.
Height/Length of Liquid	506.25 mm.
Liquid Density	0.0009874 kg./cm3

Element From Node	30
Detail Type	Nozzle
Detail ID	D1-2"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	900
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	16.892 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	30
Detail Type	Weight
Detail ID	Anti Swirl Baff
Dist. from "FROM" Node / Offset dist	-455 mm.
Miscellaneous Weight	25 Kg
Offset from Element Centerline	0 mm.



Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	SHELL
Distance "FROM" to "TO"	2350 mm.
Inside Diameter	1825 mm.
Element Thickness	48 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	48 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	63.996 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.37
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	900 mm.
Liquid Density	0.0009964 kg./cm3

Element From Node	40
Detail Type	Nozzle
Detail ID	N1-16"
Dist. from "FROM" Node / Offset dist	1455 mm.
Nozzle Diameter	16 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	1394.2 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	40
Detail Type	Nozzle
Detail ID	K1-2"
Dist. from "FROM" Node / Offset dist	1100 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	15.602 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	40
Detail Type	Nozzle
Detail ID	K2-2"
Dist. from "FROM" Node / Offset dist	150 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	15.602 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	40
Detail Type	Nozzle
Detail ID	K3-2"
Dist. from "FROM" Node / Offset dist	900 mm.



Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	15.602	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K4-2"	
Dist. from "FROM" Node / Offset dist	150	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	15.602	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K5-2"	
Dist. from "FROM" Node / Offset dist	900	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	15.602	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K6-2"	
Dist. from "FROM" Node / Offset dist	150	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	15.602	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K7-2"	
Dist. from "FROM" Node / Offset dist	900	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	15.602	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K8-2"	
Dist. from "FROM" Node / Offset dist	150	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	



Weight of Nozzle (Used if > 0)	15.602	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K9-2"	
Dist. from "FROM" Node / Offset dist	900	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	45.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	15.602	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K10-2"	
Dist. from "FROM" Node / Offset dist	150	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	45.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	15.602	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	M1-24"	
Dist. from "FROM" Node / Offset dist	1000	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	225.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	1489.6	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	

Element From Node	40	
Detail Type	Weight	
Detail ID	Demisting Weigh	
Dist. from "FROM" Node / Offset dist	2100	mm.
Miscellaneous Weight	230	Kgf
Offset from Element Centerline	0	mm.

Element From Node	40	
Detail Type	Weight	
Detail ID	Clip Weight	
Dist. from "FROM" Node / Offset dist	1175	mm.
Miscellaneous Weight	1110	Kgf
Offset from Element Centerline	0	mm.

Element From Node	40	
Detail Type	Weight	
Detail ID	Demistin Suppor	
Dist. from "FROM" Node / Offset dist	2100	mm.
Miscellaneous Weight	50	Kgf
Offset from Element Centerline	0	mm.

Element From Node	40	
Detail Type	Weight	
Detail ID	Davit Weight	
Dist. from "FROM" Node / Offset dist	1000	mm.
Miscellaneous Weight	50	Kgf
Offset from Element Centerline	1360.5	mm.

Element#	5/5
Element From Node	50
Element To Node	60
Element Type	Elliptical
Description	TOP HEAD
Distance "FROM" to "TO"	50 mm.
Inside Diameter	1825 mm.
Element Thickness	46 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	54 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	63.996 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.37
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Platform
Detail ID	SQR PLT
Dist. from "FROM" Node / Offset dist	810 mm.
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	48000 cm2
Platform Weight	2832.4 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm2
Platform Width	4000 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	2000 mm.
Platform Force Coefficient	1.2
Ladder Layout Angle	180.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	3000 mm.
Unit Weight of Ladder	0.037 Kg/mm.
Platform Length (top head platform)	4000 mm.

Element From Node	50
Detail Type	Nozzle
Detail ID	N2-16"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	16 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	460.02 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	50
Detail Type	Nozzle
Detail ID	K11-2"
Dist. from "FROM" Node / Offset dist	650 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	40.467 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	50
Detail Type	Nozzle
Detail ID	K12-2"
Dist. from "FROM" Node / Offset dist	650 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	40.467 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
	SKIRT	...	2000	...	2000
	SKIRT	...	3000	...	1000
	BOTTOM HEAD	...	3050	...	50
	SHELL	...	5400	...	2350
	TOP HEAD	...	5450	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
	SKIRT	...	10	1	1866	...	1
	SKIRT	...	10	1	1866	...	2
	BOTTOM HEAD	64.14	54	3	1825	138	3
	SHELL	64.091	48	3	1825	138	4
	TOP HEAD	64.003	54	3	1825	138	5

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
	SKIRT	...	No Calc	No Calc	10	No Calc	1
	SKIRT	...	No Calc	No Calc	10	No Calc	2
	BOTTOM HEAD	63.9963	64.6	69.2	46	45.5664	3
	SHELL	63.9963	65.8	70.4	48	46.7399	4
	TOP HEAD	63.9963	64.8	69.2	46	45.475	5

Minimum 64 69.2

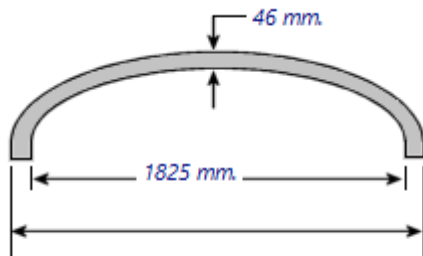
MAWP: 64 bars, limited by: DESIGN (user specified)

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: BOTTOM HEAD

Elliptical Head Design Information:					
Design Pressure	P	64.140 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	1825.000 mm.	Actual Stress		136.6 N./mm^2
Spec. Min. Thk t,e, or sa		46.0000 mm.	Spec. Nominal Thk.		54.000 mm.
Surface Area		43539.3 cm^2	SG of Contents		0.989
Element Volume		926618.9 cm3	Weight of Contents		915.9 Kg
Empty Weight		1892.0 Kg	Operating Weight		2807.9 Kg
			Comp. All. Ope. Stress		144.8 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 22.812 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 11.406 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (64.1 \cdot 1831 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 64.1)$$

$$= 42.5664 + 3.0000 = 45.5664 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.144 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 43) / (1 \cdot 1831 + 0.2 \cdot 43)$$

$$= 64.8 - 0.14 = 64.6 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 46) / (1 \cdot 1825 + 0.2 \cdot 46)$$

$$= 69.2 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (64.1 \cdot (1 \cdot 1831 + 0.2 \cdot 43)) / (2 \cdot 1 \cdot 43)$$

$$= 136.615 \text{ N./mm}^2$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (64.1 \cdot 916) / (138 \cdot 1 - 0.6 \cdot 64.1) + 3$
 $= 46.774 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.100 bars
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 1 \cdot 51) / (916 + 0.6 \cdot 51)$
 $= 74.4 - 0.1 = 74.3 \text{ bars}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (1831 / (2 \cdot 459))^2) / 6$
 $= 0.995652$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 12.009 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 46, tr = 42.5, c = 3 mm., E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.99$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -22 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 54, tr = 43.7, c = 3 mm., E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.86$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -26 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	63.996 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	46.000 mm.	Basic MDMT	-21 °C
Stress/Thk. Ratio	0.988	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

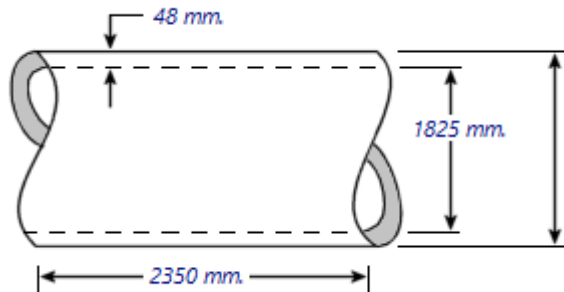
Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 22.812 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 11.406 mm.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL

Cylindrical Shell Design Information:					
Design Pressure	P	64.091 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		2350.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	1825.000 mm.	Actual Stress		134.2 N./mm^2
Spec. Min. Thk t,e, or sa		48.0000 mm.	Spec. Nominal Thk.		48.000 mm.
Surface Area		141822.5 cm^2	SG of Contents		0.998
Element Volume		6148389.0 cm3	Weight of Contents		2348.6 Kg
Empty Weight		5144.0 Kg	Operating Weight		7492.6 Kg
			Comp. All. Ope. Stress		145.3 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (64.1 \cdot 916) / (138 \cdot 1 - 0.6 \cdot 64.1)$$

$$= 43.7399 + 3.0000 = 46.7399 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.095 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 45) / (916 + 0.6 \cdot 45)$$

$$= 65.9 - 0.095 = 65.8 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 48) / (912 + 0.6 \cdot 48)$$

$$= 70.4 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (64.1 \cdot (916 + 0.6 \cdot 45)) / (1 \cdot 45)$$

$$= 134.244 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 2.563 %

Minimum Design Metal Temperature Results:

Govern. thk, tg = 48, tr = 43.7, c = 3 mm., E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.97$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -22 °C

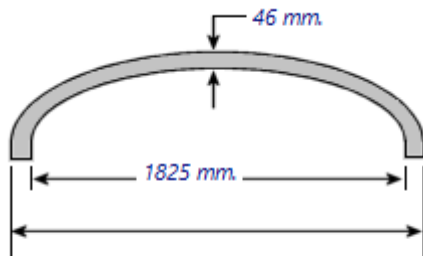
Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	63.996 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	48.000 mm.	Basic MDMT	-21 °C
Stress/Thk. Ratio	0.971	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: TOP HEAD

Elliptical Head Design Information:					
Design Pressure	P	64.003 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	1825.000 mm.	Actual Stress		136.3 N./mm^2
Spec. Min. Thk t _e , or sa		46.0000 mm.	Spec. Nominal Thk.		54.000 mm.
Surface Area		43539.3 cm^2	SG of Contents		0.000
Element Volume		926618.9 cm3	Weight of Contents		0.0 Kgf
Empty Weight		1892.0 Kgf	Operating Weight		1892.0 Kgf
			Comp. All. Ope. Stress		144.8 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 22.812 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 11.406 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (64 \cdot 1831 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 64)$$

$$= 42.4750 + 3.0000 = 45.4750 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 43) / (1 \cdot 1831 + 0.2 \cdot 43)$$

$$= 64.8 - 0.007 = 64.8 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 46) / (1 \cdot 1825 + 0.2 \cdot 46)$$

$$= 69.2 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (64 \cdot (1 \cdot 1831 + 0.2 \cdot 43)) / (2 \cdot 1 \cdot 43)$$

$$= 136.323 \text{ N./mm}^2$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (64 \cdot 916) / (138 \cdot 1 - 0.6 \cdot 64) + 3$
 $= 46.678 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.007 bars
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 1 \cdot 51) / (916 + 0.6 \cdot 51)$
 $= 74.4 - 0.007 = 74.4 \text{ bars}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (1831 / (2 \cdot 459))^2) / 6$
 $= 0.995652$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 12.009 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 46, tr = 42.5, c = 3 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.99$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -22 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 54, tr = 43.7, c = 3 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.86$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -26 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	63.996 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	46.000 mm.	Basic MDMT	-21 °C
Stress/Thk. Ratio	0.988	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 22.812 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 11.406 mm.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	83.195 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	83.195 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	89.799 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	70.396 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	91.515 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	91.515 bars
App 27-4	= M.A.W.P.	63.996 bars

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
BOTTOM HEAD	179.4	234.0	0.767	89.98
SHELL	176.5	234.0	0.754	89.98
TOP HEAD	179.4	234.0	0.767	89.98

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D1-2"	Nozzle	118.00	118.00	1.000
N1-16"	Nozzle	138.00	138.00	1.000
K1-2"	Nozzle	138.00	138.00	1.000
K2-2"	Nozzle	138.00	138.00	1.000
K3-2"	Nozzle	138.00	138.00	1.000
K4-2"	Nozzle	138.00	138.00	1.000
K5-2"	Nozzle	138.00	138.00	1.000
K6-2"	Nozzle	138.00	138.00	1.000
K7-2"	Nozzle	138.00	138.00	1.000
K8-2"	Nozzle	138.00	138.00	1.000
K9-2"	Nozzle	138.00	138.00	1.000
K10-2"	Nozzle	138.00	138.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
N2-16"	Nozzle	138.00	138.00	1.000
K11-2"	Nozzle	138.00	138.00	1.000
K12-2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
BOTTOM HEAD	138.00	138.00	1.000
SHELL	138.00	138.00	1.000
TOP HEAD	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
D1-2"	13.89	216.00	0.064
N1-16"	70.70	225.00	0.314
K1-2"	13.89	225.00	0.062
K2-2"	13.89	225.00	0.062
K3-2"	13.89	225.00	0.062
K4-2"	13.89	225.00	0.062
K5-2"	13.89	225.00	0.062



K6-2"	13.89	225.00	0.062
K7-2"	13.89	225.00	0.062
K8-2"	13.89	225.00	0.062
K9-2"	13.89	225.00	0.062
K10-2"	13.89	225.00	0.062
M1-24"	103.35	225.00	0.459
N2-16"	70.70	225.00	0.314
K11-2"	13.89	225.00	0.062
K12-2"	13.89	225.00	0.062

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	83.195 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	83.195 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	89.929 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	70.396 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	91.515 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	91.515 bars
App 27-4	= M.A.W.P.	63.996 bars

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 64 • 1
 = 83.195 bars

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
BOTTOM HEAD	177.9	234.0	0.760	83.52
SHELL	174.8	234.0	0.747	83.48
TOP HEAD	177.3	234.0	0.758	83.24

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D1-2"	Nozzle	118.00	118.00	1.000
N1-16"	Nozzle	138.00	138.00	1.000
K1-2"	Nozzle	138.00	138.00	1.000
K2-2"	Nozzle	138.00	138.00	1.000
K3-2"	Nozzle	138.00	138.00	1.000
K4-2"	Nozzle	138.00	138.00	1.000
K5-2"	Nozzle	138.00	138.00	1.000
K6-2"	Nozzle	138.00	138.00	1.000
K7-2"	Nozzle	138.00	138.00	1.000
K8-2"	Nozzle	138.00	138.00	1.000
K9-2"	Nozzle	138.00	138.00	1.000
K10-2"	Nozzle	138.00	138.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
N2-16"	Nozzle	138.00	138.00	1.000
K11-2"	Nozzle	138.00	138.00	1.000
K12-2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
BOTTOM HEAD	138.00	138.00	1.000
SHELL	138.00	138.00	1.000
TOP HEAD	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
D1-2"	12.89	216.00	0.060
N1-16"	65.59	225.00	0.292



K1-2"	12.88	225.00	0.057
K2-2"	12.88	225.00	0.057
K3-2"	12.88	225.00	0.057
K4-2"	12.88	225.00	0.057
K5-2"	12.88	225.00	0.057
K6-2"	12.88	225.00	0.057
K7-2"	12.88	225.00	0.057
K8-2"	12.88	225.00	0.057
K9-2"	12.88	225.00	0.057
K10-2"	12.88	225.00	0.057
M1-24"	95.88	225.00	0.426
N2-16"	65.41	225.00	0.291
K11-2"	12.85	225.00	0.057
K12-2"	12.85	225.00	0.057

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	3000	...	...	No Calc	No Calc
20	30	3000	...	...	No Calc	No Calc
30	40	No Calc	1917	43	0.0031154	112.565
40	50	2754.17	1921	45	0.0033217	113.646
50	60	No Calc	1917	43	0.0031154	112.565

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	46	No Calc	...	28.1
40	50	48	No Calc	...	35.5
50	60	46	No Calc	...	28.1

Minimum

28.1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	3000	No Calc	No Calc	No Calc
20	30	3000	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	2754.17	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	1418.57	...	1327.65	...	99.3
20	30	456.763	...	411.304	...	31.9734
30	40	1891.98	926619	1786.87	935355	132.439
40	50	5144.03	6148389	4830.25	6188883	360.082
50	60	1891.98	926619	1786.87	935355	132.439
Total		10803	8001627.00	10142	8059594.00	756

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Insl	1461.42	...	1000	...	Fire Proofing
20	Insl	730.709	...	500	...	Fire Proofing
30	Plat	3197.42	...	...	0.16424E-04	PLAT:[1 OF 1]
30	Liqd	915.867	...	-228.125	...	Liquid: 20
30	Nozl	18.075	...	-576.912	...	D1-2"
30	Wght	25	...	-455	...	Anti Swirl Baff
40	Liqd	2348.55	...	450	...	Liquid: 30
40	Nozl	1491.74	1037.51	1455	...	N1-16"
40	Nozl	16.6942	-992.497	1100	...	K1-2"
40	Nozl	16.6942	-992.497	150	...	K2-2"
40	Nozl	16.6942	992.497	900	...	K3-2"
40	Nozl	16.6942	992.497	150	...	K4-2"
40	Nozl	16.6942	...	900	992.497	K5-2"
40	Nozl	16.6942	...	150	992.497	K6-2"
40	Nozl	16.6942	...	900	-992.497	K7-2"
40	Nozl	16.6942	...	150	-992.497	K8-2"
40	Nozl	16.6942	701.802	900	-701.802	K9-2"
40	Nozl	16.6942	701.802	150	-701.802	K10-2"
40	Nozl	1593.87	-768.983	1000	768.983	M1-24"
40	Wght	230	...	2100	...	Demisting Weigh
40	Wght	1110	...	1175	...	Clip Weight
40	Wght	50	...	2100	...	Demistin Suppor
40	Wght	50	-962.019	1000	962.019	Davit Weight
50	Plat	3030.67	...	810	...	SQR PLT
50	Nozl	492.222	...	733.249	...	N2-16"
50	Nozl	43.3001	-650	672.223	...	K11-2"
50	Nozl	43.3001	650	672.223	...	K12-2"
40	Trun	75.5877	...	1950	...	Trunnions

Total Weight of Each Detail Type:

Platforms	6228.1
Liquid	3264.4
Insulation	2192.1
Nozzles	3849.5
Weights	1465.0
Trunnions	75.6

Sum of the Detail Weights 17074.7 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	11559.6	11559.6	11559.6	11559.6	11559.6	11559.6
Nozzles	3849.5	3849.5	3849.5	3849.5	3849.5	3849.5
Wld Weights	1235.0	1235.0	1235.0	1235.0	1235.0	1235.0
Trunnions	75.6	75.6	75.6	75.6	75.6	75.6

Platforms	...	...	...	6228.1	6228.1	6228.1
Insulation	...	...	...	2192.1	2192.1	2192.1
Empty Weights	...	...	...	230.0	230.0	...
Ope Weights	...	...	...	...	...	230.0
Ope. Liquid	...	...	...	...	...	3264.4
Test Liquid	...	7996.7	...	...	...	...
Totals	16719.6	24716.4	16719.6	25369.8	25369.8	28634.2

Field Installation Options:

Platforms installed before lifting.

Insulation installed before lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	16719.6 kg.
Shop Test Wt.	- Fabricated weight + water (full)	24716.4 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	16719.6 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	25369.8 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	25369.8 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	28634.2 kg.
Field Test Wt.	- Empty weight + test medium (full)	33136.6 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	14842.3 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	15588.7 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	14380.8 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm^2	Inside Surface Area cm^2
10	20	118501	117244
20	30	59250.4	58622.1
30	40	43539.3	38970.2
40	50	141822	134735
50	60	43539.3	38970.2

Total	406652.5	388541.8 cm2
Total	437.7	418.2 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	1517.87	2979.29	2979.29	...	2979.29
20	30	488.736	1219.45	1219.45	...	1219.45
30	40	2067.5	6180.78	6190.97	0.52515E-04	5264.92
40	50	10272.3	12620.8	16186.9	1250.35	10272.3
50	60	2603.25	5633.91	6559.97	...	5633.91

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	25369.8	28634.2	33136.6



20	30	22390.5	25655	30157.3
30	40	21171.1	24435.5	28937.8
40	50	15906.2	18254.7	22746.9
50	60	5633.91	5633.91	6559.97

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	1250.35	1250.35	1250.35
20	30	1250.35	1250.35	1250.35
30	40	1250.35	1250.35	1250.35
40	50	1250.35	1250.35	1250.35
50	60	...	...	...

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
D1-2"	143.41	153.10	82	900	GR 1.1	...	...	...	...
N1-16"	95.63	102.00	82	600	GR 1.1	...	...	...	...
K1-2"	95.63	102.00	82	600	GR 1.1	...	...	...	...
K2-2"	95.63	102.00	82	600	GR 1.1	...	...	...	...
K3-2"	95.63	102.00	82	600	GR 1.1	...	...	...	...
K4-2"	95.63	102.00	82	600	GR 1.1	...	...	...	...
K5-2"	95.63	102.00	82	600	GR 1.1	...	...	...	...
K6-2"	95.63	102.00	82	600	GR 1.1	...	...	...	...
K7-2"	95.63	102.00	82	600	GR 1.1	...	...	...	...
K8-2"	95.63	102.00	82	600	GR 1.1	...	...	...	...
K9-2"	95.63	102.00	82	600	GR 1.1	...	...	...	...
K10-2"	95.63	102.00	82	600	GR 1.1	...	...	...	...
M1-24"	95.63	102.00	82	600	GR 1.1	...	...	...	...
N2-16"	95.63	102.00	82	600	GR 1.1	27.392	95.633	95.633	95.633
K11-2"	95.63	102.00	82	600	GR 1.1	...	...	...	...
K12-2"	95.63	102.00	82	600	GR 1.1	...	...	...	...
Min Rating	95.633	102.000, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
N2-16"	438.150	0.50	4613	0	63.996	95.633	Pass

Evaluating Flange on Nozzle [N2-16"]:

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR] \\
 &= 16(4613) + 4 \cdot \text{abs}(0) \cdot 438.150 \leq \pi \cdot 438.150^3 \cdot [(95.633 - 63.996) + 0.500 \cdot 95.633] \\
 &= 0.73816E+05 \leq 0.21411E+06 \quad \text{Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle N2-16" [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(95.633, 0.500 \cdot 95.633 + 95.633 - 73815.9 / (\pi \cdot 438.150^3)) \\
 &= \min(95.633, 116.058) \\
 &= 95.633 \text{ bars}
 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 12.3677 Hz.

The Natural Frequency for the Vessel (Ope...) is 11.6657 Hz.

The Natural Frequency for the Vessel (Filled) is 10.6914 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.95) / (1 + 1.7 \cdot 3.4 \cdot 0.23)))$
 $= \min(0.85, 0.9)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	11.666 Hz
Natural Frequency of Vessel (Empty)	12.368 Hz
Natural Frequency of Vessel (Test)	10.691 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.535
Structure Height to Diameter ratio	3.129
Height to top of Structure	5952.250 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Zg = 274320 \text{ mm.}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1000 + 300 = 1300 \text{ mm.}$
 $= 4.27 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because $z (4.27 \text{ ft.}) < 15 \text{ ft.}$
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15/900)^{2/9.5}$
 $= 0.85$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1, \text{ per Table 26.6-1}$

As there is No Hill Present: $[Kzt]$:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.85 \cdot 1 \cdot 1 \cdot 1 \cdot 60.4^2)$$

$$= 16 \text{ psf } [78.1] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16 \cdot 0.85 \cdot 0.54 \cdot 58.6$$

$$= 427 \text{ lbs. } [193] \text{ Kg}$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
SKIRT	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
SKIRT	2800.0	0.000	0.000	0.000	0.849	1.000	78.120
BOTTOM HEAD	3325.0	0.000	0.000	0.000	0.849	1.000	78.120
SHELL	4525.0	0.000	0.000	0.000	0.849	1.000	78.120
TOP HEAD	5936.4	0.000	0.000	0.000	0.897	1.000	78.120

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm.
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm.
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
- L - Total height of structure mm.
- Lc - Total length of conical section(s) of vessel, mm.
- tb - Uncorroded plate thickness at bottom of vessel, mm.
- V30 - Design Wind Speed provided by user, Km/hr
- Vc - Critical wind velocity, Km/hr
- Vw - Maximum wind speed at top of structure, Km/hr
- W - Total corroded weight of structure, Kg
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kg
- Z - Maximum amplitude of vibration at top of vessel, mm.
- Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.32037E-06 (High); 0.32037E-06 < 0.40047E-06 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0/5450 = 0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[6.42/17.9^2] \cdot 10^4 = 201$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.40047E-06$ no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 27928 / (5450 \cdot 1924^2)$$

$$= 0.13842E-05$$

Since Vp is > 0.40047E-06 no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area	Elevation	Pressure	Force	Cf
----	-----------	-----------	----------	-------	----



	cm2	mm.	Kgs/m2	Kgf	
PLAT:[1 OF 1]	36000.00	3300.00	78.12	281.23	1.20
SQR PLT	48000.00	6510.00	78.12	374.98	1.20

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf
Anti Swirl Ba	0.00	2845.00	78.12	0.00
Demisting Wei	0.00	5450.00	78.12	0.00
Clip Weight	0.00	4525.00	78.12	0.00
Demistin Supp	0.00	5450.00	78.12	0.00
Davit Weight	0.00	4350.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 11.6657 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm^2	Wind Pressure Kgs/m^2	Element Wind Load Kgf
10	20	1300	2720.82	54416.4	78.12	212.836
20	30	2800	2720.82	27208.2	78.12	106.418
30	40	3325	2626.29	1313.15	78.12	312.166
40	50	4525	2631.77	61846.6	78.12	241.897
50	60	5936.37	2626.29	11673	78.12	497.988

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.300
Table Value Fv	1.500
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	C
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	No
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.3 * 0.15 = 0.19 \\
 SM1 &= Fv * S1 = 1.5 * 0.065 = 0.097 \\
 SDS &= 2/3 * Sms = 2/3 * 0.19 = 0.13 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.097 = 0.065
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_{nx}$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(19.40.75)$
 $= 0.185$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/11.7)$
 $= 0.086$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.18, 0.086)$ per 12.8.2
 $= 0.086$

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 $= Sds / (R / I)$
 $= 0.13 / (1.25 / 1.25)$
 $= 0.127$

Maximum value of C_s per equation 12.8-3 [Cs]:
 $= Sd1 / (T(R / I_e))$
 $= 0.065 / (0.086 (1.25 / 1.25))$
 $= 0.758$

Minimum value of C_s per equation 15.4-1 [Cs]:
 $= \max(0.044 \cdot SDS \cdot I_e, 0.030)$
 $= \max(0.044 \cdot 0.13 \cdot 1.25, 0.030)$
 $= 0.030$

Total Base Shear [V]:
 $= C_s \cdot W$ (Equation 12.8-1):
 $= 0.13 \cdot 28634$
 $= 3648.003$ Kgf

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} \cdot V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x \cdot h_{xk} / (\text{Sum of } W_i \cdot h_{ik})$ and k is an exponent which is related

to the period of Vibration.

In this case, the value of k was 1

The Natural Frequency for the Vessel (Ope...) is 11.6657 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	1000	2979.29	100.067	49.2756	...
20	30	2500	1219.45	102.396	39.6654	...
30	40	3025	6180.78	627.982	203.034	...
40	50	4225	12620.8	1790.99	1408.93	...
50	60	5425	5633.91	1026.57	458.472	...



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	1371.3	3648	5252.14	15609.1
20	30	2500	1158.47	3547.94	2722.32	8412.98
30	40	3025	1052.05	3445.54	1617.04	4916.17
40	50	4225	739.885	2817.56	1572.24	4759.59
50	60	5425	497.988	1026.57	117.711	242.654

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	1371.3	0.0047962	0.017466	0.15782E-04
20	30	1158.47	0.025989	0.035589	0.20096E-04
30	40	1052.05	0.036091	0.036595	0.20132E-04
40	50	739.885	0.060626	0.085115	0.20922E-04
50	60	497.988	0.085638	0.086161	0.20922E-04

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	569.787	3561.17
20	30	569.787	3561.17
30	40	549.99	3437.44
40	50	551.138	3444.61
50	60	549.99	3437.44

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 27.250 Actual deflection : 0.086 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	589.362	530.707	27495276	24785116
20	30	589.362	530.707	27495276	24785116
30	40	2703.84	2531.56	123512056	116004112
40	50	2824.41	2652.13	129033368	121541056
50	60	2703.84	2531.56	123512056	116004112



Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	115.92	196.56	-108.649	-108.649
20	30	82.8	140.4	-108.649	-108.649
30	40	165.6	280.8	-144.776	-144.776
40	50	165.6	280.8	-145.297	-145.297
50	60	165.6	280.8	-144.776	-144.776

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	...	...	...
30	40	66.8576	...	86.9053
40	50	63.8292	...	82.9689
50	60	66.8576	...	86.9053

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-4.68804	-5.29127	-6.12324	0.49472	0.49472
20	30	-4.1375	-4.74073	-5.57271	0.49472	0.49472
30	40	-0.82013	-0.82013	-0.82013	0.1057	0.1057
40	50	-0.58817	-0.58817	-0.58817	0.10089	0.10089
50	60	-0.21825	-0.21825	-0.21825	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	0.49472	2.0781	6.17598	1.03905	...
20	30	0.49472	1.07713	3.32873	0.53857	...
30	40	0.1057	0.1367	0.4156	0.06835	...
40	50	0.10089	0.12686	0.38403	0.063429	...
50	60	...	0.009951	0.020513	0.0049755	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	3.67956
20	30	...	...	...	1.99024
30	40	...	...	...	0.24852
40	50	...	...	...	0.22925
50	60	...	...	...	0.0091613

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	...	...
20	30	...	...
30	40	...	...

40	50	...	...
50	60	...	...

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm^2	Earthquake For/Mom Corroded N./mm^2	Wind For/Mom No Corrosion N./mm^2	Earthquake For/Mom No Corrosion N./mm^2
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm^2
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-7.26	108.65		0.0668
20		82.80	-5.71	108.65		0.0525
30		165.60	-1.06	144.78		0.0073
40		165.60	-0.82	145.30		0.0056
50		165.60	-0.23	144.78		0.0016

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-8.86	108.65		0.0816
20		82.80	-6.62	108.65		0.0610
30		165.60	-1.17	144.78		0.0081
40		165.60	-0.92	145.30		0.0063
50		165.60	-0.23	144.78		0.0016

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-7.86	108.65		0.0724
20		82.80	-6.31	108.65		0.0581
30		165.60	-1.06	144.78		0.0073
40		165.60	-0.82	145.30		0.0056

50 165.60 -0.23 144.78 0.0016

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.38	115.92	-11.96	108.65	0.0119	0.1101
20		82.80	-8.56	108.65		0.0788
30		165.60	-1.34	144.78		0.0093
40		165.60	-1.07	145.30		0.0074
50		165.60	-0.24	144.78		0.0016

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-7.66	108.65		0.0705
20		140.40	-6.61	108.65		0.0608
30		280.80	-0.99	144.78		0.0069
40		280.80	-0.75	145.30		0.0052
50		280.80	-0.22	144.78		0.0015

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-6.62	108.65		0.0609
20		140.40	-6.07	108.65		0.0558
30		280.80	-0.93	144.78		0.0064
40		280.80	-0.69	145.30		0.0047
50		280.80	-0.22	144.78		0.0015

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-7.86	108.65		0.0724
20		82.80	-6.31	108.65		0.0581
30	66.28	165.60		144.78	0.4002	
40	63.47	165.60		145.30	0.3833	
50	66.65	165.60		144.78	0.4025	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.38	115.92	-11.96	108.65	0.0119	0.1101
20		82.80	-8.56	108.65		0.0788
30	66.56	165.60		144.78	0.4019	
40	63.73	165.60		145.30	0.3848	
50	66.66	165.60		144.78	0.4025	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-7.86	108.65		0.0724
20		82.80	-6.31	108.65		0.0581
30		165.60	-1.06	144.78		0.0073
40		165.60	-0.82	145.30		0.0056
50		165.60	-0.23	144.78		0.0016

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.38	115.92	-11.96	108.65	0.0119	0.1101
20		82.80	-8.56	108.65		0.0788
30		165.60	-1.34	144.78		0.0093
40		165.60	-1.07	145.30		0.0074

50 165.60 -0.24 144.78 0.0016

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-7.66	108.65		0.0705
20		140.40	-6.61	108.65		0.0608
30	86.26	280.80		144.78	0.3072	
40	82.55	280.80		145.30	0.2940	
50	86.69	280.80		144.78	0.3087	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-6.62	108.65		0.0609
20		140.40	-6.07	108.65		0.0558
30	86.19	280.80		144.78	0.3069	
40	82.48	280.80		145.30	0.2937	
50	86.69	280.80		144.78	0.3087	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-5.18	108.65		0.0477
20		82.80	-4.63	108.65		0.0426
30	66.14	165.60		144.78	0.3994	
40	63.34	165.60		145.30	0.3825	
50	66.64	165.60		144.78	0.4024	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-4.76	108.65		0.0439
20		82.80	-4.27	108.65		0.0393
30	66.09	165.60		144.78	0.3991	
40	63.28	165.60		145.30	0.3821	
50	66.65	165.60		144.78	0.4025	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-5.79	108.65		0.0533
20		82.80	-5.24	108.65		0.0482
30	66.14	165.60		144.78	0.3994	
40	63.34	165.60		145.30	0.3825	
50	66.64	165.60		144.78	0.4024	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-5.18	108.65		0.0477
20		82.80	-4.63	108.65		0.0426
30	66.14	165.60		144.78	0.3994	
40	63.34	165.60		145.30	0.3825	
50	66.64	165.60		144.78	0.4024	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-5.79	108.65		0.0533
20		82.80	-5.24	108.65		0.0482
30		165.60	-0.93	144.78		0.0064
40		165.60	-0.69	145.30		0.0047

50 165.60 -0.22 144.78 0.0015

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		96.60	-5.79	90.54		0.0639
20		69.00	-5.24	90.54		0.0578
30	66.14	138.00		120.65	0.4793	
40	63.34	138.00		121.08	0.4590	
50	66.64	138.00		120.65	0.4829	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		96.60	-5.79	90.54		0.0639
20		69.00	-5.24	90.54		0.0578
30		138.00	-0.93	120.65		0.0077
40		138.00	-0.69	121.08		0.0057
50		138.00	-0.22	120.65		0.0018

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-5.79	108.65		0.0533
20		82.80	-5.24	108.65		0.0482
30	66.14	165.60		144.78	0.3994	
40	63.34	165.60		145.30	0.3825	
50	66.64	165.60		144.78	0.4024	

Absolute Maximum of the all of the Stress Ratio's 0.4829

Governing Element: TOP HEAD

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
SKIRT	...	-5.3	0.5	...	...	+ Tens
SKIRT	...	-5.3	-0.5	...	-5.8	- Comp
SKIRT	...	-4.7	0.5	...	...	+ Tens
SKIRT	...	-4.7	-0.5	...	-5.2	- Comp
BOTTOM HEAD	66.9	-0.8	0.1	...	66.1	+ Tens
BOTTOM HEAD	66.9	-0.8	-0.1	...	...	- Comp
SHELL	63.8	-0.6	0.1	...	63.3	+ Tens
SHELL	63.8	-0.6	-0.1	...	...	- Comp
TOP HEAD	66.9	-0.2	...	...	66.6	+ Tens
TOP HEAD	66.9	-0.2	...	...	...	- Comp

Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: SKIRT

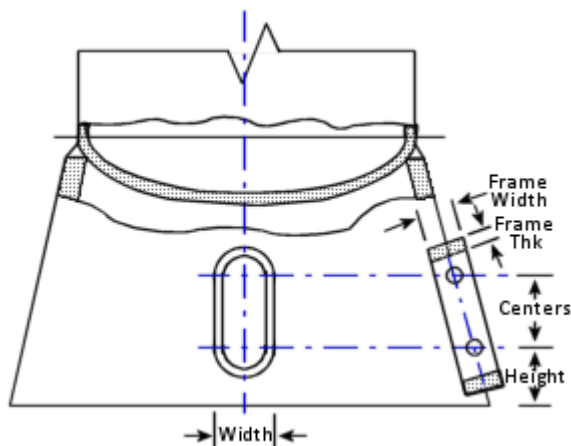
Total vessel (empty) weight	Fg	23494.492 Kgf
Head Weight below section 2-2	DeltaFg	5264.915 Kgf
Weight of vessel contents (liquid)	Ff	3264.419 Kgf
Vessel Gross Weight	Fg + Ff	26758.910 Kgf
Skirt Mean Diameter	Dz	1877.000 mm.
Skirt Analysis Thickness (Corroded)	ez	9.000 mm.
Skirt Height	H	1000.000 mm.
Additional Axial Force at section 1-1	F1	0.000 Kgf
Moment at section 1-1	M1	4916.174 Kg-m.
Applied Moment - Bottom of Skirt	Mb	8412.980 Kg-m.
Weight at Bottom of Skirt	W	25654.957 Kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	1874.000 mm.
Head Corroded Thickness	eb	43.000 mm.

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 459 \text{ mm.} \\
 K &= D_i / (2 * h_i) = 1831 / (2 * 459) = 1.99 \\
 R &= D_i (0.44 * K + 0.02) = 1831 * (0.44 * 1.99 + 0.02) = 1643 \text{ mm.} \\
 r &= D_i ((0.5 / K) - 0.08) = 1831 * ((0.5 / 1.99) - 0.08) = 313 \text{ mm.}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	2337193.750 cm ⁴
Moment of Inertia	Iyy =	2337193.750 cm ⁴
Section Modulus	W =	24903528.000 mm. ³
Section Metal Area	A =	530.709 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

$$\begin{aligned}
 &\text{Radius of Gyration of Skirt at 4-4 [rg]:} \\
 &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})} \\
 &= \sqrt{(\min(2337193, 2337193) / 531)} \\
 &= 664 \text{ mm.}
 \end{aligned}$$

Skirt Slenderness Ratio [rg/L]:

$$= r_g / L$$

$$= 664/1000$$

$$= 0.664$$

Correct the Moment [M4]:
 top of skirt is the critical end
 Controlling Moment [M4] = 4916 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:
 Downwind Side:
 $= -M / Z - W / A$
 $= -4916/99614112 - 25655/531$
 $= -5.22 \text{ N./mm}^2$

Upwind Side:
 $= +(M) / Z - W / A$
 $= +4916/99614112 - 25655/531$
 $= -4.26 \text{ N./mm}^2$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:
 $= F1 - Fg - Ff + 4 \cdot M1 / Dz$
 $= 0 - 23494 - 3264 + 4 \cdot 4916/1877$
 $= -16282 \text{ Kgf}$

Force at the joint, 16.12-2 [Fzq]:
 $= F1 - Fg - Ff - 4 \cdot M1 / Dz$
 $= 0 - 23494 - 3264 - 4 \cdot 4916/1877$
 $= -37235 \text{ Kgf}$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:
 $= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (-16282 + 5265 + 3264) / (\pi \cdot 1874 \cdot 43) + 64 \cdot 1874 / (4 \cdot 43)$
 $= 69.43 \text{ N./mm}^2$

Stress [σ_{1qm}]:
 $= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (-37235 + 5265 + 3264) / (\pi \cdot 1874 \cdot 43) + 64 \cdot 1874 / (4 \cdot 43)$
 $= 68.62 \text{ N./mm}^2$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:
 $= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (3264 + 5265) / (\pi \cdot 1874 \cdot 43) + 64 \cdot 1874 / (4 \cdot 43)$
 $= 70.06 \text{ N./mm}^2$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:
 $= Fzp / (\pi \cdot Dz \cdot ez)$
 $= -16282 / (\pi \cdot 1877 \cdot 9)$
 $= -3.01 \text{ N./mm}^2$

Stress [σ_{3qm}]:
 $= Fzq / (\pi \cdot Dz \cdot ez)$
 $= -37235 / (\pi \cdot 1877 \cdot 9)$
 $= -6.88 \text{ N./mm}^2$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [$\cos(\gamma)$]:
 $= 1 - 0.5(Db + eb - Dz + ez) / (r + eb)$
 $= 1 - 0.5(1874 + 43 - 1877 + 9) / (313 + 43)$
 $= 0.93$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.93) = 21.4 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(43^2 + 9^2 + 2 \cdot 43 \cdot 9 \cdot 0.93)}$$

$$= 25.7 \text{ mm.}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 25.74 \cdot -16282.5 = -419.16 \text{ Kg-m.}$$

$$M_q = a \cdot F_{zq} = 25.74 \cdot -37235.4 = -958.55 \text{ Kg-m.}$$

Correction factor [C]:

$$= 0.63 - 0.057 \left(\frac{eb}{ez} \right)^2$$

$$= 0.63 - 0.057 \left(\frac{43}{9} \right)^2$$

$$= -0.671$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2)$$

$$= -0.67 \cdot 6 \cdot -419 / (\pi \cdot 1874 \cdot 43^2)$$

$$= 1.52 \text{ N./mm}^2$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2)$$

$$= -0.67 \cdot 6 \cdot -959 / (\pi \cdot 1874 \cdot 43^2)$$

$$= 3.48 \text{ N./mm}^2$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2)$$

$$= -0.67 \cdot 6 \cdot -419 / (\pi \cdot 1877 \cdot 9^2)$$

$$= 34.66 \text{ N./mm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2)$$

$$= -0.67 \cdot 6 \cdot -959 / (\pi \cdot 1877 \cdot 9^2)$$

$$= 79.25 \text{ N./mm}^2$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 1.69/73.8 = 0.023$$

$$y = 125 \cdot eb/D_b = 125 \cdot 43/1874 = 2.87$$

As this an elliptical head, it will be treated as a Korbbogen type head

Geometric value [α]:

$$= 4.2 - 0.2 \cdot y = 4.2 - 0.2 \cdot 2.87 = 3.63$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) \cdot D_b \cdot (\gamma / \gamma_A) \cdot (\alpha - 1) / (4 \cdot eb)$$

$$= (64 + 0) \cdot 1874 \cdot (21.4/40) \cdot (3.63 - 1) / (4 \cdot 43)$$

$$= 65.47 \text{ N./mm}^2$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 69.4 - 1.52 + 65.5$$

$$= 133.38 \text{ N./mm}^2$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 69.4 + 1.52 - 65.5$$

$$= 5.48 \text{ N./mm}^2$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= 68.6 - 3.48 + 65.5$$



$$= 130.62 \text{ N./mm}^2$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= 68.6 + 3.48 - 65.5$$

$$= 6.62 \text{ N./mm}^2$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 70.1 + 1.52 + 65.5$$

$$= 137.06 \text{ N./mm}^2$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 70.1 - 1.52 - 65.5$$

$$= 3.07 \text{ N./mm}^2$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 70.1 + 3.48 + 65.5$$

$$= 139.01 \text{ N./mm}^2$$

Stress [σ_{2qotot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 70.1 - 3.48 - 65.5$$

$$= 1.11 \text{ N./mm}^2$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= -3.01 - 34.7$$

$$= -37.66 \text{ N./mm}^2$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= -3.01 + 34.7$$

$$= 31.65 \text{ N./mm}^2$$

Stress [σ_{3qitot}]:

$$= \sigma_{3qm} - \sigma_{3qb}$$

$$= -6.88 - 79.3$$

$$= -86.13 \text{ N./mm}^2$$

Stress [σ_{3qotot}]:

$$= \sigma_{3qm} + \sigma_{3qb}$$

$$= -6.88 + 79.3$$

$$= 72.37 \text{ N./mm}^2$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$$

$$= 138(3 - (1 / 1.5)) \cdot (69.4 / 138)^2$$

$$= 390.71 \text{ N./mm}^2$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$$

$$= 138(3 - (1 / 1.5)) \cdot (68.6 / 138)^2$$

$$= 391.25 \text{ N./mm}^2$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$$

$$= 138(3 - (1 / 1.5)) \cdot (70.1 / 138)^2$$

$$= 390.29 \text{ N./mm}^2$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$$

$$= 138(3 - (1 / 1.5)) \cdot (70.1 / 138)^2$$

$$= 390.29 \text{ N./mm}^2$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$\begin{aligned}
 &= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2) \\
 &= 138 (3 - (1/1.5) \cdot (-3.01/138)^2) \\
 &= 413.96 \text{ N./mm}^2
 \end{aligned}$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$\begin{aligned}
 &= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2) \\
 &= 138 (3 - (1/1.5) \cdot (-6.88/138)^2) \\
 &= 413.77 \text{ N./mm}^2
 \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm2)

Stress	Actual	Allowed	Result	Equation
Stot1pi	133.38	390.71	Pass	46
Stot1po	5.48	390.71	Pass	47
Stot1qi	130.62	391.25	Pass	48
Stot1qo	6.62	391.25	Pass	49
Stot2pi	137.06	390.29	Pass	50
Stot2po	3.07	390.29	Pass	51
Stot2qi	139.01	390.29	Pass	52
Stot2qo	1.11	390.29	Pass	53
Stot3pi	-37.66	413.96	Pass	54
Stot3po	31.65	413.96	Pass	55
Stot3qi	-86.13	413.77	Pass	56
Stot3qo	72.37	413.77	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.
 Insulation is installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	4562.028 mm.
Center of Gravity of the Liquid	3295.717 mm.
Center of Gravity of the Insulation	1500.000 mm.
Center of Gravity of the Nozzles	4510.824 mm.
Center of Gravity of the Added Weights (Operating)	4367.150 mm.
Center of Gravity of the Added Weights (Empty)	4367.150 mm.
Center of Gravity of the Trunnions	5000.000 mm.
Center of Gravity of Bare Shell New and Cold	3728.597 mm.
Center of Gravity of Bare Shell Corroded	3732.917 mm.
Vessel CG in the Operating Condition	3835.367 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	3967.112 mm.
Vessel CG in the Test Condition	4049.780 mm.

Rigging Analysis Results:

- Lifting weight based on the user selected calculated empty weight

Total Effective Length of Vessel for this analysis	5450.00 mm.
Total vessel weight (No Liquid) Twt	25369.83 Kgf
Impact weight multiplication factor Imp	2.00
Design lifting weight, DWT = Imp * Twt	50739.65 Kgf
Elevation of the Tailing Lug (bottom)	0.00 mm.
Elevation of the Lifting Lug (top)	5000.00 mm.
Design Reaction force at the tailing lug	10481.68 Kgf
Design Reaction force at the Trunnions	40257.98 Kgf
CG distance during field lifting operation	3967.11 mm.
CG distance from tailing lug	3967.11 mm.
CG distance from the nearer lifting lug/trunnion	1032.89 mm.

Compressive Allowable Increase Factor 1.00

Critical Values: (N./mm², mm.)

	Max Stress	Elevation	Allowables
Bending	6.90	2800.00	93.49 (UG-23)
Shear	0.93	4930.00	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kgf	Shear Stress N./mm ²
0.00	0.0	0.0	4402.3	0.7
1600.00	13084.5	4.7	767.3	0.1
2600.00	18454.4	6.6	1019.3	0.2
3520.00	18254.8	1.4	8373.6	0.3
5400.00	1250.7	0.1	779.3	0.0

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.074 at elevation 2800 mm. - Must be <=1

Note: The rigging analysis is performed using a uniformly distributed load.



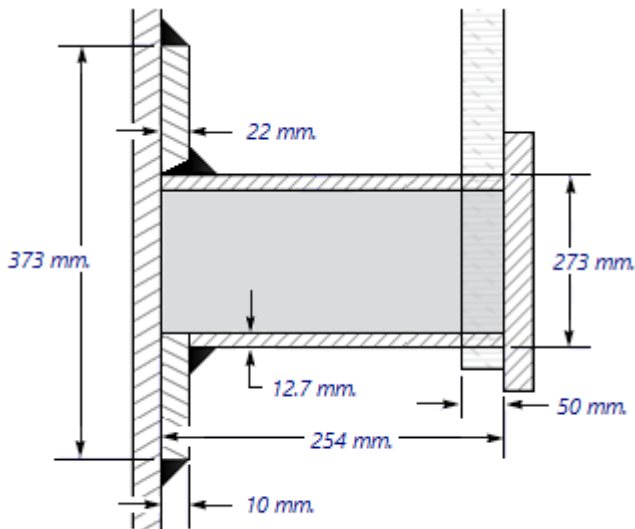
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Trunnion Analysis:

Input Data:

Lifting Lug Material	SA-106 B
Lifting Lug Yield Stress	Yield 241.00 N./mm ²
Trunnion Outside Diameter	Tod 273.000 mm.
Trunnion Thickness	Tthk 12.7000 mm.
Trunnion Length	Tl 254.0000 mm.
Reinforcing Pad Used	Yes
Pad Outside Diameter	Dp 373.000 mm.
Pad Thickness	Thkp 22.0000 mm.
Pad Edge Fillet Weld Thk	Tfp 10.0000 mm.
Bail/Sling Width	Tt 50.0000 mm.
Sling Angle from Horizontal	0.0 degrees
Occasional Load Factor	Occfac 1.0
Impact Factor	Impfac 2.00
Number of Lugs in Group	2



Trunnion may be inserted or abutting

Trunnion Force and Stress Results for different Lift Angles (Kgf&N/mm²):

Bending Stress Allowable:	159.1 N./mm ²
Shear Stress Allowable:	96.4 N./mm ²
Bearing Stress Allowable:	180.7 N./mm ²

Angle	Trunnion Lifting Load	Bending Stress	Shear Stress	Bearing Stress	Unity Check	Pass Fail
0	20129.0	70.0	19.0	14.5	0.637	Pass
5	20213.2	70.3	19.1	14.5	0.640	Pass
10	20295.9	70.6	19.2	14.6	0.642	Pass
15	20378.5	70.9	19.2	14.6	0.645	Pass
20	20462.3	71.1	19.3	14.7	0.648	Pass
25	20548.5	71.4	19.4	14.8	0.650	Pass
30	20638.7	71.8	19.5	14.8	0.653	Pass
35	20734.6	72.1	19.6	14.9	0.656	Pass
40	20838.5	72.5	19.7	15.0	0.660	Pass
45	20953.1	72.8	19.8	15.1	0.663	Pass
50	21082.4	73.3	19.9	15.1	0.667	Pass
55	21231.8	73.8	20.0	15.3	0.672	Pass
60	21409.1	74.4	20.2	15.4	0.678	Pass
65	21626.8	75.2	20.4	15.5	0.685	Pass
70	21905.2	76.2	20.7	15.7	0.693	Pass
75	22280.4	77.5	21.0	16.0	0.705	Pass
80	22823.6	79.4	21.6	16.4	0.722	Pass



85	23697.0	82.4	22.4	17.0	0.750	Pass
90	25369.8	88.2	24.0	18.2	0.803	Pass

Trunnion Calculation Results:

Trunnion Lifting Load [Tlf]:
 = Resultant Load • Impact Factor
 = 12685 • 2
 = 25369.8 Kgf

Trunnion Section Modulus [Z]:
 = $\pi (T_{od}^4 - T_{id}^4) / (32 \cdot T_{od})$
 = $\pi (273^4 - 248^4) / (32 \cdot 273)$
 = 645930.9 mm.³

Trunnion Bending Moment [Tmom]:
 = $(T_l - 0.5 \cdot T_t) / 9.807 \cdot T_{lf}$
 = $(254 - 0.5 \cdot 50) / 9.807 \cdot 25370$
 = 5809.8 Kg-m.

Trunnion Calculated Bending Stress [Sbc]:
 = Tmom • 9.807/Z
 = 5810 • 9.807/645931
 = 88.21 N./mm [Passed] Sbc < min(0.66 • SY • Occfac, 0.75 • Sy) (159.06)

Trunnion Cross Section Area [Txsa]:
 = $\pi/4 (T_{od}^2 - T_{id}^2)$
 = $\pi/4 (273^2 - 248^2)$
 = 103.9 cm²

Trunnion Calculated Shear Stress [Ssc]:
 = Tlf/Txsa
 = 25370/104
 = 23.96 N./mm [Passed] Ssc < 0.4 • Sy • Occfac (96.40)

Trunnion Unity Check [Uchk]:
 = Sbc/Sba + Ssc/Ssa
 = 88.2/159 + 24/96.4
 = 0.803 [Passed]

Weld Section Modulus at Edge of Trunnion [Sw]:
 = $\pi (T_{od}/2)^2$
 = $\pi (273/2)^2$
 = 58534.9 mm.²

Shear Stress in Weld due to Bending Moment [fs]:
 = Tmom/Sw
 = 5810/58535
 = 99.252 Kgf/mm.

Required Fillet Weld Leg Size at Edge of Trunnion [Wsr]:
 = max(10mm, fs/(0.4 • Sy))
 = max(10mm, 99.3/(0.4 • 241))
 = 10.097 mm.

Trunnion Bearing Contact Area [Tbca]:
 = T_{od} • T_t
 = 273 • 50
 = 136.5 cm²

Trunnion Bearing Stress [Sbec]:
 = Tlf/Tbca
 = 25370/136
 = 18.23 N./mm [Passed] Sbec < min(0.75 • SY • Occfac, 0.9 • Sy) (180.75)

Summary of Results:

Trunnion Unity Check Value: 0.8031 Must be less than or equal to 1

Comparison of Stress Values versus Allowables: (N/mm²)

Type of Stress	Actual	Allowable	Pass/Fail
Bending Stress	88.21	159.06	Pass
Shear Stress	23.96	96.40	Pass
Bearing Stress	18.23	180.75	Pass

Results of local shell stress analysis due to loads on the trunnion:

This following Stress Analysis assumes that the Trunnion passes through the pad and is welded to the shell via a groove weld. Additionally the re-pad is welded to the shell at the outside edge via a fillet weld.

In the local stress results below, the 'Nozzle' is the 'Trunnion'.

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).

Loads on shell and WRC 107 results at different Lift Angles (Kgf & Kg-m.):

Angle	PT	PL	MC	ML	Rat1	Rat2	P/F
0	20129.0	0.0	4609.6	0.0	0.117	0.162	Pass
3	20152.1	1056.1	4614.9	241.9	0.117	0.162	Pass
5	20136.3	1761.7	4611.3	403.4	0.117	0.162	Pass
10	19987.6	3524.4	4577.2	807.1	0.116	0.161	Pass
15	19684.2	5274.4	4507.8	1207.9	0.115	0.159	Pass
20	19228.3	6998.5	4403.4	1602.7	0.112	0.155	Pass
25	18623.3	8684.2	4264.8	1988.7	0.109	0.151	Pass
30	17873.6	10319.3	4093.1	2363.2	0.105	0.145	Pass
35	16984.8	11892.9	3889.6	2723.5	0.101	0.138	Pass
40	15963.2	13394.7	3655.6	3067.5	0.096	0.131	Pass
45	14816.1	14816.1	3393.0	3393.0	0.090	0.122	Pass
50	13551.5	16150.1	3103.4	3698.4	0.085	0.113	Pass
55	12178.0	17392.0	2788.8	3982.9	0.090	0.103	Pass
60	10704.5	18540.8	2451.4	4245.9	0.095	0.108	Pass
65	9139.9	19600.5	2093.1	4488.6	0.099	0.113	Pass
70	7492.0	20584.2	1715.7	4713.9	0.103	0.117	Pass
75	5766.6	21521.2	1320.6	4928.5	0.107	0.122	Pass
80	3963.3	22476.8	907.6	5147.3	0.112	0.127	Pass
85	2065.3	23606.8	473.0	5406.1	0.117	0.133	Pass
90	0.0	25369.8	0.0	5809.8	0.125	0.142	Pass

- PT - Transverse load per lug
- PL - Longitudinal load per lug
- MC - Circumferential moment in shell
- ML - Longitudinal moment in shell
- Rat1 - Computed WRC 107 stress ratio at trunnion edge
- Rat2 - Computed WRC 107 stress ratio at pad edge

Worst case combination loads were at: 3.0 degrees from horizontal.

Input Echo, WRC107/537 Item 1, Description: Trunnion

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	1825.000 mm.
Vessel Thickness	Tv	48.000 mm.
Design Temperature	T1	37.8 C
Attachment Type	Type	Round
Diameter Basis for Nozzle	Nbasis	OD
Corrosion Allowance for Nozzle	Can	0.0000 mm.
Nozzle Diameter	Dn	273.000 mm.
Nozzle Thickness	Tn	12.700 mm.

Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm2
Thickness of Reinforcing Pad	Tpad	22.000	mm.
Diameter of Reinforcing Pad	Dpad	393.000	mm.
Design Internal Pressure	Dp	0.000	bars
Include Pressure Thrust		No	

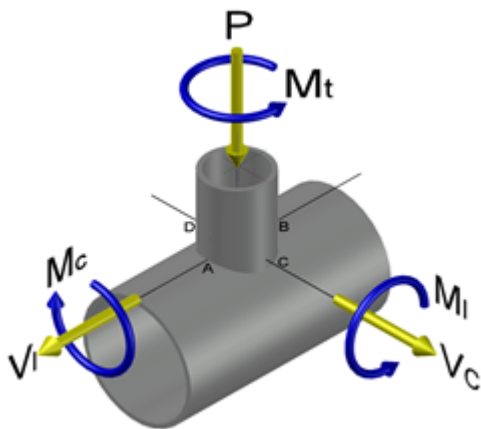
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	0.0	Kgf
Longitudinal Shear (SUS)	VL	1056.1	Kgf
Circumferential Shear (SUS)	Vc	20152.1	Kgf
Circumferential Moment (SUS)	Mc	4614.9	Kg-m.
Longitudinal Moment (SUS)	ML	241.9	Kg-m.
Torsional Moment (SUS)	Mt	0.0	Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



Worst case combination loads were at: 3.0 degrees from horizontal.

WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 273.0 + 2 \cdot 1.65 \cdot \sqrt{936.5(48.0 - 0)}$$

$$= 972.662 \text{ mm.}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	20152.1	Kgf
Longitudinal Shear	VL	1056.1	Kgf
Circumferential Moment	MC	4614.9	Kg-m.
Longitudinal Moment	ML	241.9	Kg-m.
Torsional Moment	MT	0.0	Kg-m.

Dimensionless Parameters used : Gamma = 13.54

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
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N(PHI) / (P/Rm)	0.126	4C	2.510	(A,B)
N(PHI) / (P/Rm)	0.126	3C	2.366	(C,D)
M(PHI) / (P)	0.126	2C1	0.112	(A,B)
M(PHI) / (P)	0.126	1C	0.145	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.126	3A	0.314	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.126	1A	0.101	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.126	3B	1.144	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.126	1B	0.053	(A,B,C,D)
N(x) / (P/Rm)	0.126	3C	2.366	(A,B)
N(x) / (P/Rm)	0.126	4C	2.510	(C,D)
M(x) / (P)	0.126	1C1	0.149	(A,B)
M(x) / (P)	0.126	2C	0.113	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.126	4A	0.435	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.126	2A	0.058	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.126	4B	0.297	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.126	2B	0.085	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-1.8	-1.8	1.8	1.8
Circ. Bend. MC		0.0	0.0	0.0	0.0	-46.7	46.7	46.7	-46.7
Circ. Memb. ML		-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Circ. Bend. ML		-1.3	1.3	1.3	-1.3	0.0	0.0	0.0	0.0
Tot. Circ. Str.		-1.6	0.9	1.6	-0.9	-48.5	44.9	48.5	-44.9
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-2.5	-2.5	2.5	2.5
Long. Bend. MC		0.0	0.0	0.0	0.0	-26.8	26.8	26.8	-26.8
Long. Memb. ML		-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Long. Bend. ML		-2.1	2.1	2.1	-2.1	0.0	0.0	0.0	0.0
Tot. Long. Str.		-2.2	2.0	2.2	-2.0	-29.3	24.4	29.3	-24.4
Shear VC		6.6	6.6	-6.6	-6.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		6.6	6.6	-6.6	-6.6	-0.3	-0.3	0.3	0.3
Str. Int.		13.2	13.2	13.2	13.2	48.5	44.9	48.5	44.9

Dimensionless Parameters used : Gamma = 19.51

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.184	4C	3.150	(A,B)
N(PHI) / (P/Rm)	0.184	3C	2.496	(C,D)
M(PHI) / (P)	0.184	2C1	0.065	(A,B)
M(PHI) / (P)	0.184	1C	0.095	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.184	3A	0.717	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.184	1A	0.092	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.184	3B	2.171	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.184	1B	0.040	(A,B,C,D)
N(x) / (P/Rm)	0.184	3C	2.496	(A,B)

N(x)	/ (P/Rm)	0.184	4C	3.150	(C,D)
M(x)	/ (P)	0.184	1C1	0.099	(A,B)
M(x)	/ (P)	0.184	2C	0.065	(C,D)
N(x)	/ (MC/(Rm**2 * Beta))	0.184	4A	1.113	(A,B,C,D)
M(x)	/ (MC/(Rm * Beta))	0.184	2A	0.049	(A,B,C,D)
N(x)	/ (ML/(Rm**2 * Beta))	0.184	4B	0.681	(A,B,C,D)
M(x)	/ (ML/(Rm * Beta))	0.184	2B	0.062	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-4.2	-4.2	4.2	4.2
Circ. Bend. MC		0.0	0.0	0.0	0.0	-63.0	63.0	63.0	-63.0
Circ. Memb. ML		-0.7	-0.7	0.7	0.7	0.0	0.0	0.0	0.0
Circ. Bend. ML		-1.4	1.4	1.4	-1.4	0.0	0.0	0.0	0.0
Tot. Circ. Str.		-2.1	0.8	2.1	-0.8	-67.2	58.8	67.2	-58.8
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-6.5	-6.5	6.5	6.5
Long. Bend. MC		0.0	0.0	0.0	0.0	-33.4	33.4	33.4	-33.4
Long. Memb. ML		-0.2	-0.2	0.2	0.2	0.0	0.0	0.0	0.0
Long. Bend. ML		-2.2	2.2	2.2	-2.2	0.0	0.0	0.0	0.0
Tot. Long. Str.		-2.5	2.0	2.5	-2.0	-39.9	26.9	39.9	-26.9
Shear VC		6.7	6.7	-6.7	-6.7	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		6.7	6.7	-6.7	-6.7	-0.3	-0.3	0.3	0.3
Str. Int.		13.3	13.4	13.3	13.4	67.2	58.8	67.2	58.8

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		-0.3	-0.3	0.3	0.3	-1.8	-1.8	1.8	1.8
Circ. Q (SUS)		-1.3	1.3	1.3	-1.3	-46.7	46.7	46.7	-46.7
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		-0.1	-0.1	0.1	0.1	-2.5	-2.5	2.5	2.5
Long. Q (SUS)		-2.1	2.1	2.1	-2.1	-26.8	26.8	26.8	-26.8
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		6.6	6.6	-6.6	-6.6	-0.3	-0.3	0.3	0.3
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		13.2	13.2	13.2	13.2	2.6	2.6	2.6	2.6
Pm+Pl+Q (Total)		13.2	13.2	13.2	13.2	48.5	44.9	48.5	44.9



Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	13.17	207.00	Passed
Pm+Pl+Q (TOTAL)	48.51	414.00	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress Load	Stress Intensity Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)	-0.7	-0.7	0.7	0.7	-4.2	-4.2	4.2	4.2
Circ. Q (SUS)	-1.4	1.4	1.4	-1.4	-63.0	63.0	63.0	-63.0
Long. Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)	-0.2	-0.2	0.2	0.2	-6.5	-6.5	6.5	6.5
Long. Q (SUS)	-2.2	2.2	2.2	-2.2	-33.4	33.4	33.4	-33.4
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	6.7	6.7	-6.7	-6.7	-0.3	-0.3	0.3	0.3
Shear Q (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)	13.3	13.3	13.3	13.3	6.6	6.6	6.6	6.6
Pm+Pl+Q (Total)	13.3	13.4	13.3	13.4	67.2	58.8	67.2	58.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	13.35	207.00	Passed
Pm+Pl+Q (TOTAL)	67.24	414.00	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	1886.0000	mm.
Skirt Thickness	STHK	10.0000	mm.
Skirt Internal Corrosion Allowance	SCA	1.0000	mm.
Skirt External Corrosion Allowance		0.0000	mm.
Skirt Material		SA-516 70	[Normalized]

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	25.0000	mm.
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	138.00	N./mm2
Basing Yield Stress		262.00	N./mm2
Inside Diameter of Basing	DI, D4	1742.0000	mm.
Outside Diameter of Basing	DOU, D5	2142.0000	mm.
Nominal Diameter of Bolts	BND, d(b0)	36.0000	mm.
Bolt Corrosion Allowance	BCA	0.0000	mm.
Root Area of a Single Bolt	Area	6.1100	cm2
Bolt Material		SA-307 A	
Bolt Material Class/Thickness Range		6 <= t <= 100	
Bolt Operating Allowable Stress	SA, fB	138.00	N./mm2
Number of Bolts	RN, nb	12	
Diameter of Bolt Circle	DC, DBC	2013.9999	mm.
Bolt Allowable Shear Stress		82.800	N./mm2
Thickness of Gusset Plates	TGA, e7	16.0000	mm.
Width of Gussets at Top Plate	TWDT	118.0000	mm.
Width of Gussets at Base Plate	BWDT	118.0000	mm.
Gusset Plate Elastic Modulus	E	201734000.0	KPa.
Gusset Plate Yield Stress	SY	262.0	N./mm2
Height of Chair	h1	250.0000	mm.
Distance between Gussets	RG, b6	76.0000	mm.
Dist. from Bolt Center to Gusset (Rg/2)	CG	38.0000	mm.
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	32.0000	mm.
Radial Width of the Top Plate	TOPWTH, b5	128.0000	mm.
Anchor Bolt Hole Dia. in Top Plate	dia,H	50.0000	mm.
External Corrosion Allowance	CA	1.0000	mm.
Dead Weight of Vessel	DW	25369.8	Kgf
Operating Weight of Vessel	ROW	28634.2	Kgf
Earthquake Moment on Basing	EQMOM	15657.2	Kg-m.
Wind Moment on Basing	WIMOM	5300.2	Kg-m.
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Basing Analysis : Analyze Option

Basing Thickness Calculation method used : Simplified (Steel on Steel)

Calculation of Bolt Loads for Multiple Loading Conditions:

Bolt Load, Earthquake Operating Case [FB,e]:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 15657/2014) - 28634) / 12$$

$$= 205.1 \text{ Kgf}$$

Bolt Load, Windload Empty Case [FB,w]:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 5300/2014) - 25370) / 12$$

= -1236.9 Kgf

Bolt Load, Windload Vortex Shedding Case [FB,v]:

= ((4 • M/DC) - W)/nb per Jawad & Farr, Eq. 12.3

= ((4 • 0/2014) - 28634)/12

= -2386.2 Kgf

Calculation of Maximum Load per Bolt [W/Bolt], Earthquake + Operating Condition:

where: W = ROW - Yforce (from gy Acceleration or user force), M = EQMOM

= ((4 • M/DC) - W)/nb per Jawad & Farr, Eq. 12.3

= ((4 • 15657/2014) - 28634)/12

= 205.1 Kgf

Required Area for Each Bolt, Based on Max Load	0.1458	cm ²
Area Available in a Single Bolt (Corr)	6.1100	cm ²
Area Available in all the Bolts (Corr)	73.3200	cm ²
Bolt Stress Based on Simplified Analysis	3.3	N./mm ²
Allowable Bolt Stress 138.0 [Fact]	138.00	N./mm ²

Shear Stress in a Single Bolt [tb]:

= Shear Force/(Bolt Area • Number of Bolts)

= 3648/(6.11 • 12)

= 4.9 N./mm². Must be less than 82.8 N./mm².

Concrete Contact Area of Base Ring	CCA	12201.95	cm ²
Concrete Contact Section Modulus of Base Ring		0.5428E+09	mm. ³

Concrete Load (Simplified method), Earthquake in Operating Condition [Sc]:

= ((ppl/100•(Abt•Sa)+W)/Cca) + M/CZ per Jawad & Farr Eq. 12.1

= (1 (73.3 • 138 + 28634)/12202) + 15657/542787840

= 1.34 N./mm²

Concrete Allowable Stress	22.50	N./mm ²
---------------------------	-------	--------------------

Determine Maximum Bending Width of Basing Section [Rw1]:

= (Dou - SkirtOD)/2

= (2142 - 1886)/2

= 128 mm.

Determine Maximum Bending Width of Basing Section [Rw2]:

= (SkirtID - Di + 2•Sca)/2

= (1866 - 1742 + 2•1)/2

= 63 mm.

Calculation of required Basing Thickness, (Simplified) [Tb]:

= max(Rw1,Rw2) • √(3 • Sc / S) + CA per Jawad & Farr Eq. 12.12

= max(128,63) • √(3 • 1.34/175) + 1

= 20.4350 mm.

Basing Stress at given Thickness [Sb]:

= 3 • Sc(max[Rw1, Rw2]/(Tb - Ca))²

= 3 • 1.34 (max[128, 63]/(25 - 1))²

= 115, must be less than 175 N./mm²

Required Thickness of Top Plate in Tension:

(Calculated as a fixed beam per Megyesy)

Ft = (Sa•Abss), Bolt Allowable Stress • Area

Rm = (Ft • 2 • Cg)/8, Bending Moment

Sb Allowable Bending Stress

Wt = (Topwth - Bnd), Width of Section

Required Thickness of Top Plate [T]:

= √(6 • Rm / (Sb • Wt)) + CA

= √(6 • 81.7/(175 • 92)) + 1

= 18.294 mm.

Required Thickness of Continuous Top Ring per Moss:

a = (Dc-SkirtOD)/2 Skirt Distance to Bolt Circle

P = Sa•Abss Bolt Allowable Stress • Area



l = Avgwdt Average Gusset Width
 $g1$ = Gamma 1 Constant Term $f(b/l)$
 $g2$ = Gamma 2 Constant Term $f(b/l)$
 g = Flat distance / 2 Nut 1/2 Dimension (from Tema)
 Fb = Allowable Bending Stress
 $M_o = P/(4\pi)[1.3(\ln((2l\sin(\pi a/l)/(\pi g))) + 1) - [(0.7 - g2)P/(4\pi)]$ Moment Term

Required Thickness of Top Ring [T]:
 $= \sqrt{(6 \cdot Abs(M_o) / F_b)} + CA$ Required Thickness
 $= \sqrt{(6 \cdot 26.8/175)} + 1$
 = 19.838 mm.

Required Thickness of Gusset in Compression, per AISC E2-1:

1. Allowed Compression at Given Thickness:
 Factor Kl/r Per E2-1 41.7850
 Factor C_c Per E2-1 123.2879
 Allowable Buckling Str. per E2-1 138.04 N./mm²
 Actual Buckling Str. at Given Thickness 22.33 N./mm²

 Required Gusset/Support Plate thickness, + CA 6.3897 mm.

 2. Allowed Compression at Calculated Thickness:
 Factor Kl/r Per E2-1 124.0441
 Factor C_c Per E2-1 123.2879
 Allowable Buckling Str. per E2-2 67.52 N./mm²
 Act. Buckling Str. at Calculated Thickness 66.29 N./mm²

Summary of Basing Thickness Calculations:

Required Basing Thickness (simplified) 20.4350 mm.
 Actual Basing Thickness 25.0000 mm.

 Required Top Ring/Plate Thickness as a Fixed Beam 18.2943 mm.
 Required Thickness of Continuous Top Ring (Moss) 19.8376 mm.
 Actual Top Plate/Ring Thickness 32.0000 mm.

 Required Gusset/Support Plate thickness, + CA 6.3897 mm.
 Actual Gusset/Support Plate Thickness 16.0000 mm.

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:
 $= SkirtStress \cdot (SkirtThickness - CA)$
 $= 12 \cdot (10 - 1)$
 = 11 Kg/mm.

Results for Computed Minimum Basing Weld Size [BWeld]:
 $= W / [(0.4 \cdot Yield) \cdot 2 \cdot 0.707]$
 $= 11 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707]$
 = 0.78 mm.

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:
 $= Bolt Load / (Cmwth + 2 \cdot (Hg + Tta))$
 $= 8598 / (108 + 2 \cdot (193 + 32))$
 = 15.4 Kg/mm.

Horizontal Plate Load [Wh]:
 $= Bolt Load \cdot e / (Cmwth \cdot (Hg + Tta) + 0.6667 \cdot (Hg + Tta)^2)$
 $= 8598 \cdot 64 / (108 \cdot (225) + 0.6667 \cdot (225)^2)$
 = 9.48 Kg/mm.

Resultant Weld Load [Wr]:
 $= \sqrt{ Wv^2 + Wh^2 }$
 $= \sqrt{ 15.4^2 + 9.48^2 }$
 = 18.1 Kg/mm.

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:
 $= Wr / [(0.4 \cdot Yield) \cdot 2 \cdot 0.707]$



$$= 18.1 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707]$$

$$= 1.29 \text{ mm.}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:
 = Bolt Load / (2 • TopWth)
 = 8598 / (2 • 118)
 = 36.4 Kgf/mm.

Weld Load [Wh]:
 = Bolt Load • e / (2 • Hgt • TopWth)
 = 8598 • 64 / (2 • 225 • 118)
 = 10.4 Kgf/mm.

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{36.4^2 + 10.4^2}$$
 = 37.9 Kgf/mm.

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:
 = Wr / [(0.4 • Yield) • 2 • 0.707]
 = 37.9 / [(0.4 • 243) • 2 • 0.707]
 = 2.71 mm.

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	4.7625	mm.
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm.
Required Top Plate to Skirt Weld Size	6.3500	mm.
Required Gusset to Top Plate Double Fillet Weld Size	2.7071	mm.

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: D1-2"

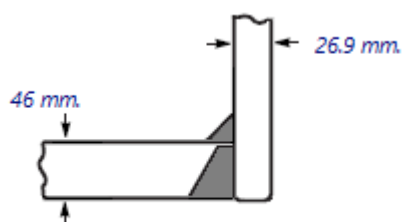
From: 30

Pressure for Reinforcement Calculations	P	64.140	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	1825.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	46.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	187.3250	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	46.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		900	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		0.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		1210.0	mm.
Centerline Distance to Tangent		800.0	mm.

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: D1-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 1.060 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 64.1 \cdot 1831 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 64.1)$
 $= 38.3570$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([64.1 / (118 \cdot 1)] - 1))$
 $= 1.5866$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	190.6480	mm.
Parallel to Vessel Wall	Rn+tn+t	95.3240	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	59.8100	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

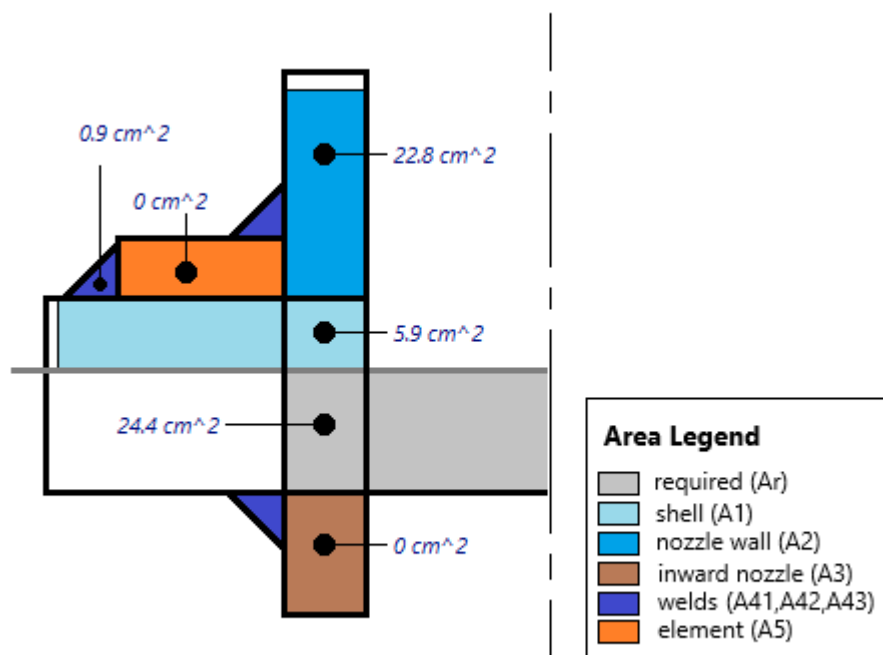


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	24.447	NA	NA
Area in Shell	A1	5.893	NA	NA
Area in Nozzle Wall	A2	22.847	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	29.595	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (56.8 \cdot 38.4 \cdot 1 + 2 \cdot 23.9 \cdot 38.4 \cdot 1 \cdot (1 - 0.86))$
 $= 24.447 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 134 (1 \cdot 43 - 1 \cdot 38.4) - 2 \cdot 23.9$
 $(1 \cdot 43 - 1 \cdot 38.4) \cdot (1 - 0.86)$
 $= 5.893 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 59.8) (23.9 - 1.59) 0.86$
 $= 22.847 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$
 $= 0.855 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 4.5866 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 45.5664 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 45.5664 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 8.2578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.26, \max(45.6, 4.5)]$
 $= 8.2578 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(4.59, 8.26)$
 $= 8.2578 \text{ mm.}$

Available Nozzle Neck Thickness = 26.9240 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	12.1,	Allowable	:	118.0 N./mm2	Passed
Expansion	:	0.0,	Allowable	:	282.9 N./mm2	Passed
Occasional	:	2.7,	Allowable	:	156.9 N./mm2	Passed
Shear	:	8.8,	Allowable	:	82.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 26.9, tr = 1.59, c = 3 mm., E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.066$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 1 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $64.14/153.10 = 0.419$

Weld Size Calculations, Description: D1-2"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 \cdot Wo$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (24.4 - 5.89 + 2 \cdot 23.9 \cdot 0.86 \cdot (1 \cdot 43 - 38.4)) \cdot 138)$
 $= 28782.52 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (22.8 + 0 + 0.86 - 0 \cdot 0.86) \cdot 138$
 $= 33353.34 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (22.8 + 0 + 0.86 + (17.6)) \cdot 138$
 $= 58109.19 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (22.8 + 0 + 0.86 + 0 + (17.6)) \cdot 138$
 $= 58109.19 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 105 \cdot 10 \cdot 0.49 \cdot 118$
 $= 9692. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 40.4) \cdot (26.9 - 3) \cdot 0.7 \cdot 118$
 $= 25551. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 105 \cdot (46 - 3) \cdot 0.74 \cdot 138$
 $= 73604. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (9692 + 25551) = 35242 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (9692 + 0 + 73604 + 0) = 83296 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (9692 + 73604 + 0) = 83296 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 35242	Kgf, must exceed W = 28782 Kg	or W1 = 33353 Kg
Path 2-2 = 83295	Kgf, must exceed W = 28782 Kg	or W2 = 58109 Kg
Path 3-3 = 83295	Kgf, must exceed W = 28782 Kg	or W3 = 58109 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 64 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.8155 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 234.1404 mm.

Input Echo, WRC107/537 Item 1, Description: D1-2" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	3285.000	mm.
Vessel Thickness	Tv	46.000	mm.
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	50.800	mm.
Nozzle Thickness	Tn	26.924	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²
Design Internal Pressure	Dp	64.140	bars
Include Pressure Thrust		No	

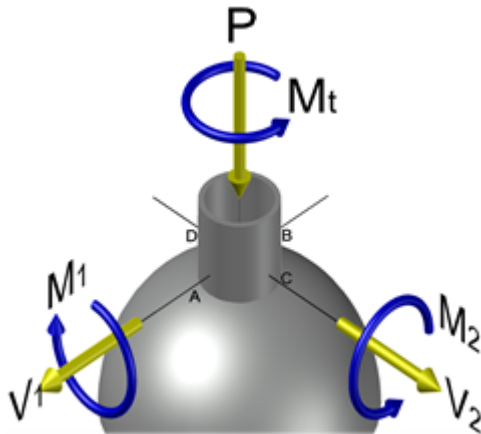
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	535.2	Kgf
Longitudinal Shear	(SUS)	(V1) V1	407.8	Kgf
Circumferential Shear	(SUS)	(Vc) V2	407.8	Kgf
Circumferential Moment	(SUS)	(Mc) M1	63.7	Kg-m.
Longitudinal Moment	(SUS)	(Ml) M2	63.7	Kg-m.
Torsional Moment	(SUS)	Mt	76.5	Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 104.648 + 2 \cdot 1.65 \cdot \sqrt{(1667.0 (46.0 - 3.0))}$$

$$= 988.167 \text{ mm.}$$

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	535.2	Kgf
Circumferential Shear	(VC) V2	407.8	Kgf
Longitudinal Shear	(VL) V1	407.8	Kgf
Circumferential Moment	(MC) M1	63.7	Kg-m.
Longitudinal Moment	(ML) M2	63.7	Kg-m.
Torsional Moment	MT	76.5	Kg-m.

Dimensionless Param: U = 0.20 TAU = 5.00 (1.69) RHO = 1.80

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 2	0.13737	(A,B,C,D)
$M(x) / P$	SP 2	0.12555	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.40845	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 2	0.45435	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.40845	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 2	0.45435	(A,B,C,D)
$N(y) * T / P$	SP 2	0.22016	(A,B,C,D)
$M(y) / P$	SP 2	0.10316	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.07216	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 2	0.54874	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.07216	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 2	0.54874	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Stress Intensity Values at

Type of Stress	Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4
Rad.	Bend. P	-2.1	2.1	-2.1	2.1	-2.1	2.1	-2.1	2.1
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-3.4	3.4	3.4	-3.4
Rad.	Memb. ML	-0.5	-0.5	0.5	0.5	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-3.4	3.4	3.4	-3.4	0.0	0.0	0.0	0.0
Tot. Rad.	Str.	-6.5	4.7	1.4	-1.2	-6.5	4.7	1.4	-1.2
Tang.	Memb. P	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6
Tang.	Bend. P	-1.8	1.8	-1.8	1.8	-1.8	1.8	-1.8	1.8
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-4.2	4.2	4.2	-4.2
Tang.	Memb. ML	-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-4.2	4.2	4.2	-4.2	0.0	0.0	0.0	0.0
Tot. Tang.	Str.	-6.6	5.2	1.9	-2.9	-6.6	5.2	1.9	-2.9
	Shear VC	0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
	Shear VL	0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
	Shear MT	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	Tot. Shear	1.6	1.6	0.4	0.4	0.4	0.4	1.6	1.6
	Str. Int.	8.1	6.5	2.1	3.0	7.0	5.5	3.2	3.9

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	122.8	122.8	122.8	122.8	122.8	122.8	122.8	122.8
Rad.	Pl (SUS)	-0.9	-0.9	0.1	0.1	-0.9	-0.9	0.1	0.1
Rad.	Q (SUS)	-5.6	5.6	1.3	-1.3	-5.6	5.6	1.3	-1.3
Long.	Pm (SUS)	122.8	122.8	122.8	122.8	122.8	122.8	122.8	122.8
Long.	Pl (SUS)	-0.7	-0.7	-0.5	-0.5	-0.7	-0.7	-0.5	-0.5
Long.	Q (SUS)	-5.9	5.9	2.4	-2.4	-5.9	5.9	2.4	-2.4
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.6	0.6	-0.6	-0.6	-0.6	-0.6	0.6	0.6
Shear	Q (SUS)	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	Pm (SUS)	122.8	122.8	122.8	122.8	122.8	122.8	122.8	122.8
	Pm+Pl (SUS)	122.5	122.5	123.2	123.2	122.5	122.5	123.2	123.2
	Pm+Pl+Q (Total)	117.8	129.3	124.9	121.7	116.7	128.2	126.0	122.5

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	122.79	138.00	Passed
Pm+Pl (SUS)	123.24	207.00	Passed
Pm+Pl+Q (TOTAL)	129.32	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N1-16"

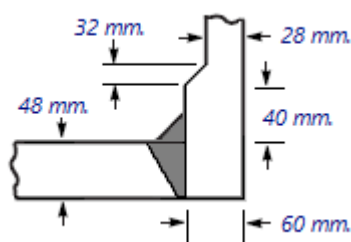
From: 40

Pressure for Reinforcement Calculations	P	64.140	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1825.00	mm.
Shell Finished (Minimum) Thickness	t	48.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1505.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2 +IT	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	28.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	40.0000	mm.
Height of Beveled Transition	L'	32.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		48.0000	mm.
Inside Projection	h	1600.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: N1-16"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 16.000 in.
 Actual Thickness Used in Calculation 1.102 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 64.1 \cdot 916 / (138 \cdot 1 - 0.6 \cdot 64.1)$
 $= 43.7744$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 64.1 \cdot 206 / (138 \cdot 1 - 0.6 \cdot 64.1)$
 $= 9.8594$ mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 32 / 0.5773$
 $= 55.4256$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

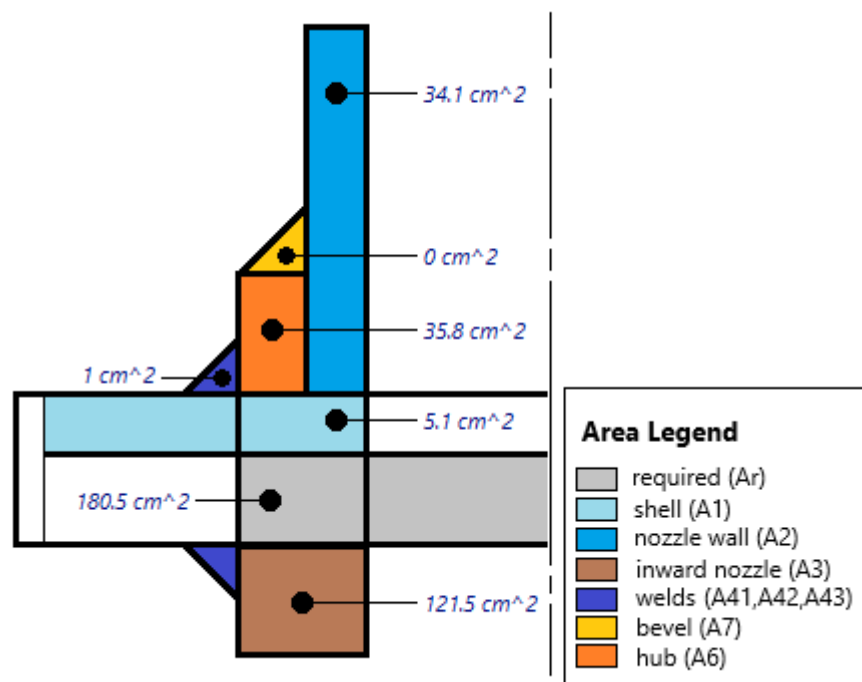
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	824.8000	mm.
Parallel to Vessel Wall, opening length	d	412.4000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	112.5000	mm.
Normal to Vessel Wall, Inward		55.0000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	180.526	NA	NA
Area in Shell	A1	5.054	NA	NA
Area in Nozzle Wall	A2	34.066	NA	NA
Area in Inward Nozzle	A3	121.500	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	35.840	NA	NA
TOTAL AREA AVAILABLE	Atot	197.461	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (412 \cdot 43.8 \cdot 1 + 2 \cdot 57 \cdot 43.8 \cdot 1 \cdot (1 - 1))$$

$$= 180.526 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 412 (1 \cdot 45 - 1 \cdot 43.8) - 2 \cdot 57 (1 \cdot 45 - 1 \cdot 43.8) \cdot (1 - 1)$$

$$= 5.054 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 113) (25 - 9.86) \cdot 1$$

$$= 34.066 \text{ cm}^2$$

Area Available in Inward Nozzle [A3]:

$$= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$$

$$= 2 \cdot 54 \cdot (113) \cdot 1$$

$$= 121.500 \text{ cm}^2$$



Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{hp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(113, 250, 40)) \cdot (60 - 28) \cdot 1$$

$$= 35.840 \text{ cm}^2 \text{ Includes Bevel Area of } 10.240 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 12.8594 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 46.7744 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 46.7744 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(46.8, 4.5)]$$

$$= 11.3312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(12.9, 11.3)$$

$$= 12.8594 \text{ mm.}$$

Available Nozzle Neck Thickness = 28.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 38.7,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 306.3 N./mm ²	Passed
Occasional	: 24.9,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 7.8,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govern. thk, tg = 28, tr = 9.86, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.39, Temp. Reduction = 52 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-30 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govern. thk, tg = 48, tr = 43.8, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.97, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-22 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -22 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 64.14/102.00 = 0.629

Weld Size Calculations, Description: N1-16"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (181 - 5.05 + 2 \cdot 57 \cdot 1 \cdot (1 \cdot 45 - 43.8)) \cdot 138)$
 $= 248886.23 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (69.9 + 1 - 0 \cdot 1) \cdot 138$
 $= 99778.16 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (69.9 + 122 + 1 + (51.3)) \cdot 138$
 $= 342939.31 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (69.9 + 122 + 1 + 0 + (51.3)) \cdot 138$
 $= 342939.31 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 526 \cdot 10 \cdot 0.49 \cdot 138$
 $= 57014. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 235) \cdot (60 - 3) \cdot 0.7 \cdot 138$
 $= 413987. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 526 \cdot (48 - 3) \cdot 0.74 \cdot 138$
 $= 387463. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (57014 + 413987) = 471001 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (57014 + 0 + 387463 + 0) = 444477 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (57014 + 387463 + 0) = 444477 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 471000 Kgf, must exceed W = 248886 Kgf	or W1 = 99778 Kgf
Path 2-2 = 444477 Kgf, must exceed W = 248886 Kgf	or W2 = 342939 Kgf
Path 3-3 = 444477 Kgf, must exceed W = 248886 Kgf	or W3 = 342939 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 64 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 38.7826 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1898.0000 mm.

Input Echo, WRC297 Item 1, Description: N1-16" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1825.000	mm.
Shell Thickness	48.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	406.400	mm.
Nozzle Thickness	60.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	64.14	bars
Radial Load	P	3669.81	Kgf
Circumferential Shear	Vc	2650.42	Kgf
Longitudinal Shear	Vl	2650.42	Kgf
Circumferential Moment	Mc	3262.23	Kg-m.
Longitudinal Moment	Ml	3262.23	Kg-m.
Torsional Moment	Mt	3751.55	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (8.773) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 526.4 + 2 \cdot 1.65 \cdot \sqrt{938.0(48.0 - 3.0)}$$

$$= 1204.387 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1876.000	mm.
Nozzle Outside Diameter	(d) =	526.400	mm.
Vessel Thickness used	(T) =	45.000	mm.
Nozzle Thickness used	(t) =	57.000	mm.
T / t	=	0.789	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.812$$

Nr/P	=	0.083
Mr/P	=	0.136
M0/P	=	0.042
N0/P	=	0.104
MrD/Mc	=	0.268
NrDL/Mc	=	0.125
M0d/Mc	=	0.091
N0DL/Mc	=	0.059
MrD/M1	=	0.123
NrDL/M1	=	0.074
M0D/M1	=	0.043
N0DL/M1	=	0.132

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-1	-1	-1	-1
Outplane Bending (P)	-4	4	-4	4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-7	7	7	-7
Normal Pressure Stress	127	133	127	133
Outplane Stress Summary	109	140	132	132

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-14	14	-14	14
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-22	22	22	-22
Inplane Pressure Stress	63	63	63	63
Inplane Stress Summary	23	96	72	56

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	109	140	132	132

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-1	-1	-1	-1



Outplane Bending (P)	-4	4	-4	4
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-16	16	16	-16
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63	63	63	63

Outplane Stress Summary	39	80	75	51
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-14	14	-14	14
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-48	48	48	-48
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	127	133	127	133

Inplane Stress Summary	59	191	163	102
------------------------	----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	2	2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	59	191	163	102
------------------------	----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-1	-1	-1	-1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	24	32	24	32

Hoop Stress Summary	18	26	26	34
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-6	6	-6	6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-9	9	9	-9
Axial Pressure Stress	12	12	12	12

Axial Stress Summary -7 24 18 11

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	2	2	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	26	27	26	34

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-1	-1	-1	-1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	24	32	24	32
Hoop Stress Summary	20	28	24	32

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-6	6	-6	6
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-23	23	23	-23
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	12	12	12	12
Axial Stress Summary	-21	37	32	-1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	2	2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	41	37	32	33

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int.	Stress Values at
------------------------	------------------



Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	127	133	127	133	127	133	127	133
Circ. Pl (SUS)	-4	-4	2	2	-4	-4	2	2
Circ. Q (SUS)	-11	11	3	-3	-62	62	34	-34
Long. Pm (SUS)	63	63	63	63	63	63	63	63
Long. Pl (SUS)	-3	-3	1	1	-2	-2	0	0
Long. Q (SUS)	-36	36	8	-8	-20	20	12	-12
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	127.0	133.0	127.0	133.0	127.0	133.0	127.0	133.0
Pm+Pl (SUS)	123.0	129.0	129.0	135.0	123.0	129.0	129.0	135.0
Pm+Pl+Q (Total)	112.0	140.0	132.0	132.0	61.0	191.0	163.0	101.0

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	133.00	138.00	Passed
Pm+Pl (SUS)	135.00	207.00	Passed
Pm+Pl+Q (TOTAL)	191.01	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm2):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	24	32	24	32	24	32	24	32
Circ. Pl (SUS)	-4	-4	2	2	-2	-2	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	12	12	12	12	12	12	12	12
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	-15	15	3	-3	-29	29	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	24.0	32.0	24.0	32.0	24.0	32.0	24.0	32.0
Pm+Pl (SUS)	20.0	28.0	26.0	34.0	22.0	30.0	24.0	32.0
Pm+Pl+Q (Total)	26.1	28.2	26.1	34.0	42.0	38.1	32.1	34.1

Nozzle Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	32.00	138.00	Passed
Pm+Pl (SUS)	34.00	207.00	Passed
Pm+Pl+Q (TOTAL)	42.05	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: K1-2"

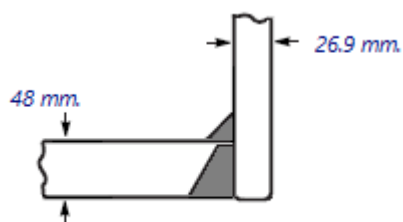
From: 40

Pressure for Reinforcement Calculations	P	64.140	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1825.00	mm.
Shell Finished (Minimum) Thickness	t	48.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1150.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	48.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K1-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	1.060	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 64.1 \cdot 916 / (138 \cdot 1 - 0.6 \cdot 64.1)$
 $= 43.7744$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([64.1 / (138 \cdot 1)] - 1))$
 $= 1.3513$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 194.6480 mm.
 Parallel to Vessel Wall Rn+tn+t 97.3240 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 59.8100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

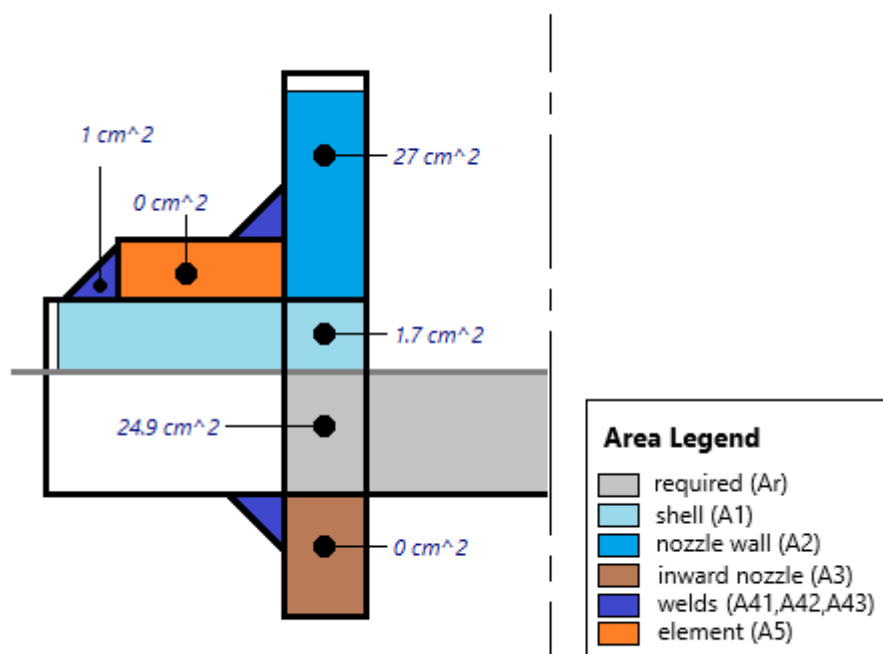


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	24.864	NA	NA
Area in Shell	A1	1.689	NA	NA
Area in Nozzle Wall	A2	27.001	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	29.690	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 43.8 \cdot 1 + 2 \cdot 23.9 \cdot 43.8 \cdot 1 \cdot (1 - 1))$
 $= 24.864 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 138(1 \cdot 45 - 1 \cdot 43.8) - 2 \cdot 23.9$
 $(1 \cdot 45 - 1 \cdot 43.8) \cdot (1 - 1)$
 $= 1.689 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 59.8)(23.9 - 1.35)1$
 $= 27.001 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 4.3513 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 46.7744 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 46.7744 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 8.2578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.26, \max(46.8, 4.5)]$
 $= 8.2578 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(4.35, 8.26)$
 $= 8.2578 \text{ mm.}$

Available Nozzle Neck Thickness = 26.9240 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:D)

Govrn. thk, $t_g = 26.9$, $tr = 1.35$, $c = 3 \text{ mm.}$, $E* = 1$
 Thickness Ratio $= tr \cdot E* / (t_g - c) = 0.056$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $64.14/102.00 = 0.629$

Weld Size Calculations, Description: K1-2"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (24.9 - 1.69 + 2 \cdot 23.9 \cdot 1 \cdot (1 \cdot 45 - 43.8)) \cdot 138)$
 $= 33436.35 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (27 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 39402.54 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (27 + 0 + 1 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (27 + 0 + 1 + 0 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 105 \cdot 10 \cdot 0.49 \cdot 138$
 $= 11334. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 40.4) \cdot (26.9 - 3) \cdot 0.7 \cdot 138$
 $= 29881. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 105 \cdot (48 - 3) \cdot 0.74 \cdot 138$
 $= 77027. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (11334 + 29881) = 41215 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 0 + 77027 + 0) = 88362 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 77027 + 0) = 88362 \text{ Kgf}$

Summary of Failure Path Calculations:



Path 1-1 = 41215 Kgf, must exceed W = 33436 Kgf or W1 = 39402 Kgf
Path 2-2 = 88361 Kgf, must exceed W = 33436 Kgf or W2 = 69700 Kgf
Path 3-3 = 88361 Kgf, must exceed W = 33436 Kgf or W3 = 69700 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 64 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.5014 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 209.5013 mm.

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Input, Nozzle Desc: K2-2"

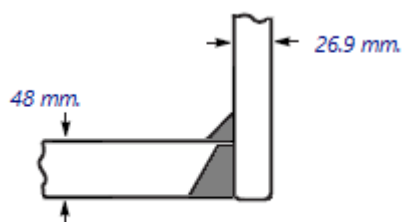
From: 40

Pressure for Reinforcement Calculations	P	64.140	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1825.00	mm.
Shell Finished (Minimum) Thickness	t	48.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		200.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	48.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K2-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	1.060	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 64.1 \cdot 916 / (138 \cdot 1 - 0.6 \cdot 64.1)$
 $= 43.7744$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([64.1 / (138 \cdot 1)] - 1))$
 $= 1.3513$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 194.6480 mm.
 Parallel to Vessel Wall Rn+tn+t 97.3240 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 59.8100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

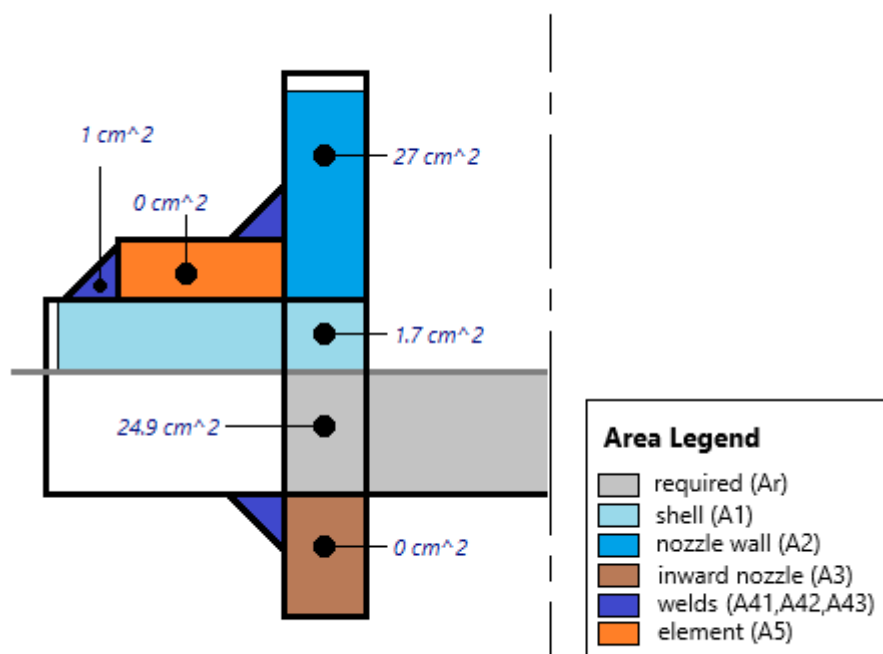


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	24.864	NA	NA
Area in Shell	A1	1.689	NA	NA
Area in Nozzle Wall	A2	27.001	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA

Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	29.690	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 43.8 \cdot 1 + 2 \cdot 23.9 \cdot 43.8 \cdot 1 \cdot (1 - 1))$
 $= 24.864 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 138(1 \cdot 45 - 1 \cdot 43.8) - 2 \cdot 23.9$
 $(1 \cdot 45 - 1 \cdot 43.8) \cdot (1 - 1)$
 $= 1.689 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 59.8)(23.9 - 1.35)1$
 $= 27.001 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 4.3513 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 46.7744 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 46.7744 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 8.2578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.26, \max(46.8, 4.5)]$
 $= 8.2578 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(4.35, 8.26)$
 $= 8.2578 \text{ mm.}$

Available Nozzle Neck Thickness = 26.9240 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:D)

Govrn. thk, $t_g = 26.9$, $tr = 1.35$, $c = 3 \text{ mm.}$, $E* = 1$
 Thickness Ratio $= tr \cdot E* / (t_g - c) = 0.056$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $64.14/102.00 = 0.629$

Weld Size Calculations, Description: K2-2"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (24.9 - 1.69 + 2 \cdot 23.9 \cdot 1 \cdot (1 \cdot 45 - 43.8)) \cdot 138)$
 $= 33436.35 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (27 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 39402.54 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (27 + 0 + 1 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (27 + 0 + 1 + 0 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 105 \cdot 10 \cdot 0.49 \cdot 138$
 $= 11334. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 40.4) \cdot (26.9 - 3) \cdot 0.7 \cdot 138$
 $= 29881. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 105 \cdot (48 - 3) \cdot 0.74 \cdot 138$
 $= 77027. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (11334 + 29881) = 41215 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (11334 + 0 + 77027 + 0) = 88362 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (11334 + 77027 + 0) = 88362 \text{ Kgf}$

Summary of Failure Path Calculations:



Path 1-1 = 41215 Kgf, must exceed W = 33436 Kgf or W1 = 39402 Kgf
Path 2-2 = 88361 Kgf, must exceed W = 33436 Kgf or W2 = 69700 Kgf
Path 3-3 = 88361 Kgf, must exceed W = 33436 Kgf or W3 = 69700 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 64 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.5014 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 209.5013 mm.

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Input, Nozzle Desc: K3-2"

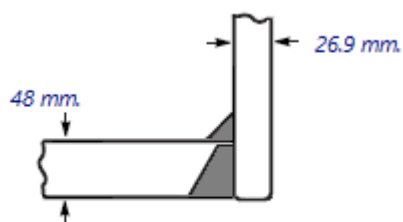
From: 40

Pressure for Reinforcement Calculations	P	64.140	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1825.00	mm.
Shell Finished (Minimum) Thickness	t	48.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		950.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	48.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K3-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	1.060	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 64.1 \cdot 916 / (138 \cdot 1 - 0.6 \cdot 64.1)$
 $= 43.7744$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([64.1 / (138 \cdot 1)] - 1))$
 $= 1.3513$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 194.6480 mm.
 Parallel to Vessel Wall Rn+tn+t 97.3240 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 59.8100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

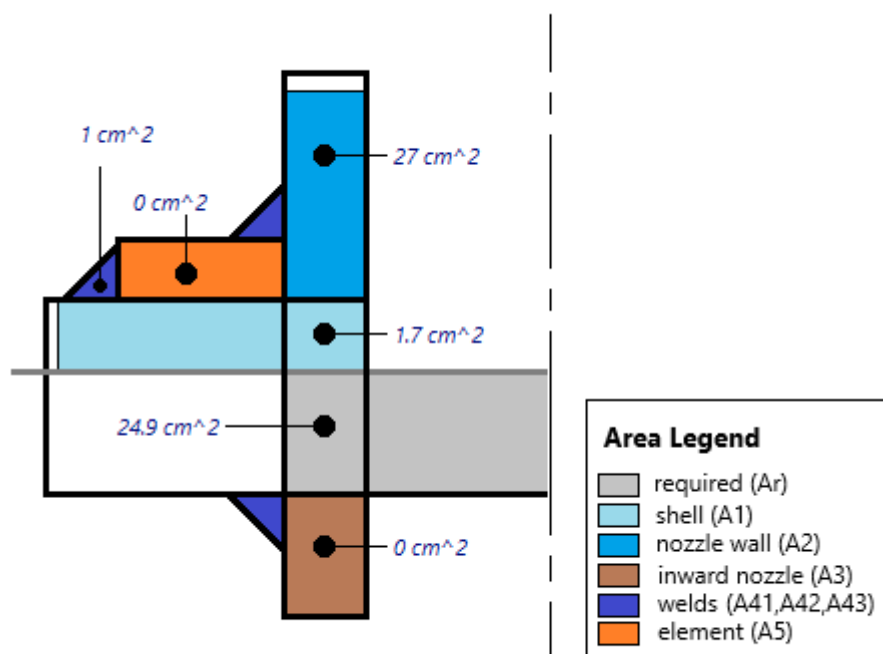


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	24.864	NA	NA
Area in Shell	A1	1.689	NA	NA
Area in Nozzle Wall	A2	27.001	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	29.690	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 43.8 \cdot 1 + 2 \cdot 23.9 \cdot 43.8 \cdot 1 \cdot (1 - 1))$
 $= 24.864 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 138(1 \cdot 45 - 1 \cdot 43.8) - 2 \cdot 23.9$
 $(1 \cdot 45 - 1 \cdot 43.8) \cdot (1 - 1)$
 $= 1.689 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 59.8)(23.9 - 1.35)1$
 $= 27.001 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 4.3513 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 46.7744 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 46.7744 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 8.2578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.26, \max(46.8, 4.5)]$
 $= 8.2578 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(4.35, 8.26)$
 $= 8.2578 \text{ mm.}$

Available Nozzle Neck Thickness = 26.9240 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:D)

Govrn. thk, $t_g = 26.9$, $tr = 1.35$, $c = 3 \text{ mm.}$, $E* = 1$
 Thickness Ratio $= tr \cdot E* / (t_g - c) = 0.056$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $64.14/102.00 = 0.629$

Weld Size Calculations, Description: K3-2"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (24.9 - 1.69 + 2 \cdot 23.9 \cdot 1 \cdot (1 \cdot 45 - 43.8)) \cdot 138)$
 $= 33436.35 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (27 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 39402.54 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (27 + 0 + 1 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (27 + 0 + 1 + 0 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 105 \cdot 10 \cdot 0.49 \cdot 138$
 $= 11334. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 40.4) \cdot (26.9 - 3) \cdot 0.7 \cdot 138$
 $= 29881. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 105 \cdot (48 - 3) \cdot 0.74 \cdot 138$
 $= 77027. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (11334 + 29881) = 41215 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 0 + 77027 + 0) = 88362 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 77027 + 0) = 88362 \text{ Kgf}$

Summary of Failure Path Calculations:



Path 1-1 = 41215 Kgf, must exceed W = 33436 Kgf or W1 = 39402 Kgf
Path 2-2 = 88361 Kgf, must exceed W = 33436 Kgf or W2 = 69700 Kgf
Path 3-3 = 88361 Kgf, must exceed W = 33436 Kgf or W3 = 69700 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 64 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.5014 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 209.5013 mm.

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Input, Nozzle Desc: K4-2"

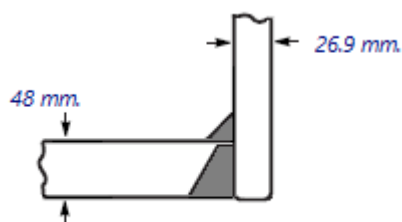
From: 40

Pressure for Reinforcement Calculations	P	64.140	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1825.00	mm.
Shell Finished (Minimum) Thickness	t	48.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		200.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	48.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K4-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	1.060	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 64.1 \cdot 916 / (138 \cdot 1 - 0.6 \cdot 64.1)$
 $= 43.7744$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([64.1 / (138 \cdot 1)] - 1))$
 $= 1.3513$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 194.6480 mm.
 Parallel to Vessel Wall Rn+tn+t 97.3240 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 59.8100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

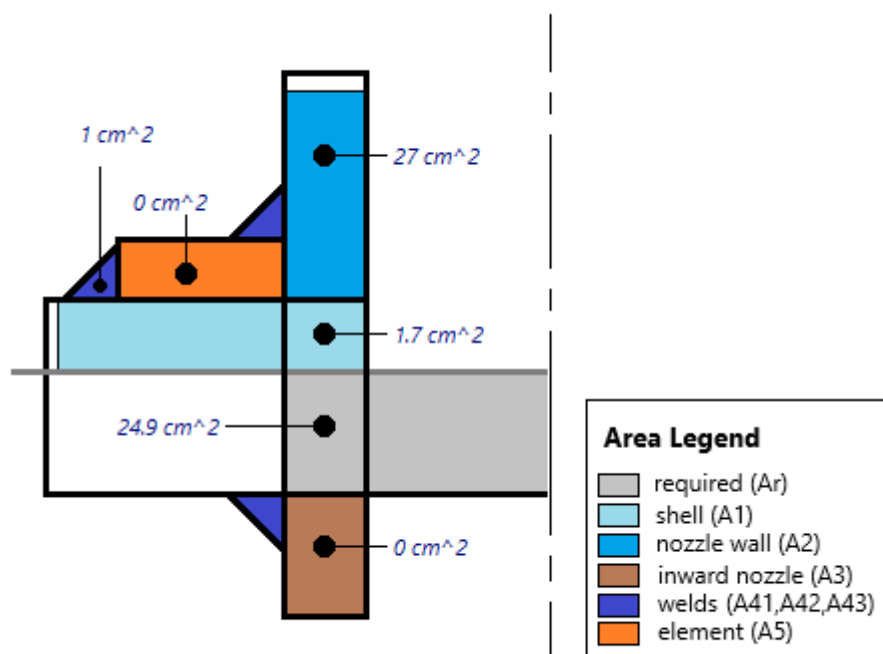


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	24.864	NA	NA
Area in Shell	A1	1.689	NA	NA
Area in Nozzle Wall	A2	27.001	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	29.690	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (56.8 \cdot 43.8 \cdot 1 + 2 \cdot 23.9 \cdot 43.8 \cdot 1 \cdot (1 - 1))$$

$$= 24.864 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 138(1 \cdot 45 - 1 \cdot 43.8) - 2 \cdot 23.9$$

$$(1 \cdot 45 - 1 \cdot 43.8) \cdot (1 - 1)$$

$$= 1.689 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 59.8)(23.9 - 1.35)1$$

$$= 27.001 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 4.3513 \text{ mm.}$

Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm.}$

Wall Thickness, shell/head, internal pressure $trb1 = 46.7744 \text{ mm.}$

Wall Thickness $tb1 = \max(trb1, tr16b) = 46.7744 \text{ mm.}$

Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$

Wall Thickness per table UG-45 $tb3 = 8.2578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.26, \max(46.8, 4.5)]$$

$$= 8.2578 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.35, 8.26)$$

$$= 8.2578 \text{ mm.}$$

Available Nozzle Neck Thickness = 26.9240 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:D)

Govrn. thk, $t_g = 26.9$, $tr = 1.35$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.056$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $64.14/102.00 = 0.629$

Weld Size Calculations, Description: K4-2"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (24.9 - 1.69 + 2 \cdot 23.9 \cdot 1 \cdot (1 \cdot 45 - 43.8)) \cdot 138)$
 $= 33436.35 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (27 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 39402.54 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (27 + 0 + 1 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (27 + 0 + 1 + 0 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 105 \cdot 10 \cdot 0.49 \cdot 138$
 $= 11334. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 40.4) \cdot (26.9 - 3) \cdot 0.7 \cdot 138$
 $= 29881. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 105 \cdot (48 - 3) \cdot 0.74 \cdot 138$
 $= 77027. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (11334 + 29881) = 41215 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 0 + 77027 + 0) = 88362 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 77027 + 0) = 88362 \text{ Kgf}$

Summary of Failure Path Calculations:



Path 1-1 = 41215 Kgf, must exceed W = 33436 Kgf or W1 = 39402 Kgf
Path 2-2 = 88361 Kgf, must exceed W = 33436 Kgf or W2 = 69700 Kgf
Path 3-3 = 88361 Kgf, must exceed W = 33436 Kgf or W3 = 69700 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 64 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.5014 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 209.5013 mm.

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Input, Nozzle Desc: K5-2"

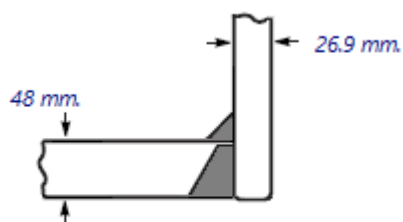
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Pressure for Reinforcement Calculations	P	64.140	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1825.00	mm.
Shell Finished (Minimum) Thickness	t	48.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		950.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	48.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K5-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	1.060	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 64.1 \cdot 916 / (138 \cdot 1 - 0.6 \cdot 64.1)$
 $= 43.7744$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([64.1 / (138 \cdot 1)] - 1))$
 $= 1.3513$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 194.6480 mm.
 Parallel to Vessel Wall Rn+tn+t 97.3240 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 59.8100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

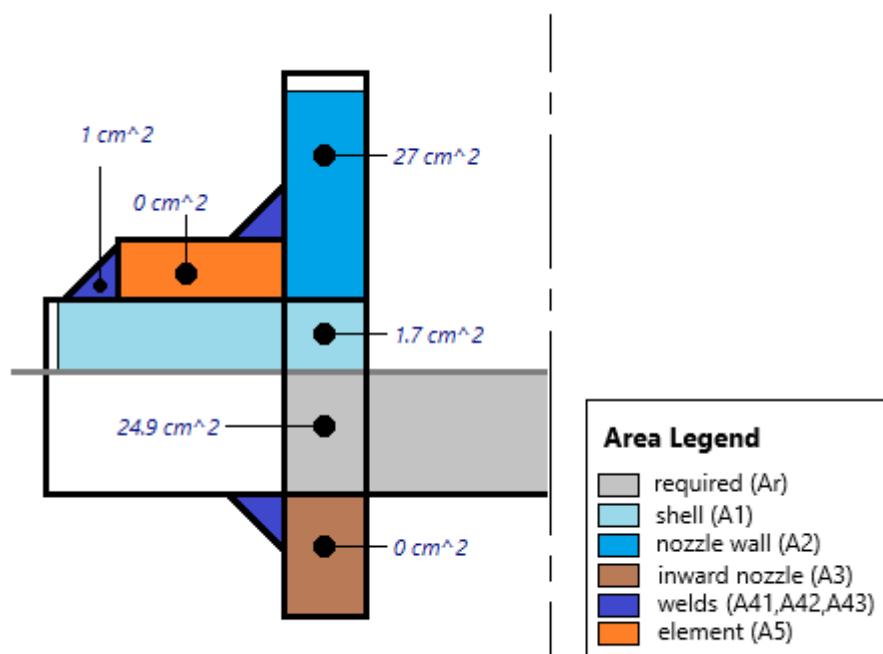


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	24.864	NA	NA
Area in Shell	A1	1.689	NA	NA
Area in Nozzle Wall	A2	27.001	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	29.690	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 43.8 \cdot 1 + 2 \cdot 23.9 \cdot 43.8 \cdot 1 \cdot (1 - 1))$
 $= 24.864 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 138(1 \cdot 45 - 1 \cdot 43.8) - 2 \cdot 23.9$
 $(1 \cdot 45 - 1 \cdot 43.8) \cdot (1 - 1)$
 $= 1.689 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2$
 $= (2 \cdot 59.8)(23.9 - 1.35)1$
 $= 27.001 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 4.3513 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 46.7744 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 46.7744 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 8.2578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.26, \max(46.8, 4.5)]$
 $= 8.2578 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(4.35, 8.26)$
 $= 8.2578 \text{ mm.}$

Available Nozzle Neck Thickness = 26.9240 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:D)

Govrn. thk, $t_g = 26.9$, $tr = 1.35$, $c = 3 \text{ mm.}$, $E* = 1$
 Thickness Ratio $= tr \cdot E* / (t_g - c) = 0.056$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $64.14/102.00 = 0.629$

Weld Size Calculations, Description: K5-2"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (24.9 - 1.69 + 2 \cdot 23.9 \cdot 1 \cdot (1 \cdot 45 - 43.8)) \cdot 138)$
 $= 33436.35 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (27 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 39402.54 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (27 + 0 + 1 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (27 + 0 + 1 + 0 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 105 \cdot 10 \cdot 0.49 \cdot 138$
 $= 11334. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 40.4) \cdot (26.9 - 3) \cdot 0.7 \cdot 138$
 $= 29881. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 105 \cdot (48 - 3) \cdot 0.74 \cdot 138$
 $= 77027. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (11334 + 29881) = 41215 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 0 + 77027 + 0) = 88362 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 77027 + 0) = 88362 \text{ Kgf}$

Summary of Failure Path Calculations:



Path 1-1 = 41215 Kgf, must exceed W = 33436 Kgf or W1 = 39402 Kgf
Path 2-2 = 88361 Kgf, must exceed W = 33436 Kgf or W2 = 69700 Kgf
Path 3-3 = 88361 Kgf, must exceed W = 33436 Kgf or W3 = 69700 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 64 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.5014 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 209.5013 mm.

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Input, Nozzle Desc: K6-2"

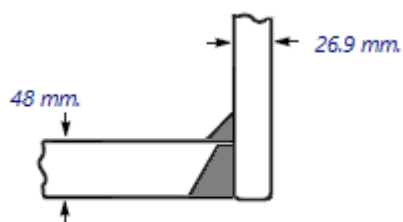
From: 40

Pressure for Reinforcement Calculations	P	64.140	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1825.00	mm.
Shell Finished (Minimum) Thickness	t	48.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		200.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	48.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K6-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	1.060	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 64.1 \cdot 916 / (138 \cdot 1 - 0.6 \cdot 64.1)$
 $= 43.7744$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([64.1 / (138 \cdot 1)] - 1))$
 $= 1.3513$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 194.6480 mm.
 Parallel to Vessel Wall Rn+tn+t 97.3240 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 59.8100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

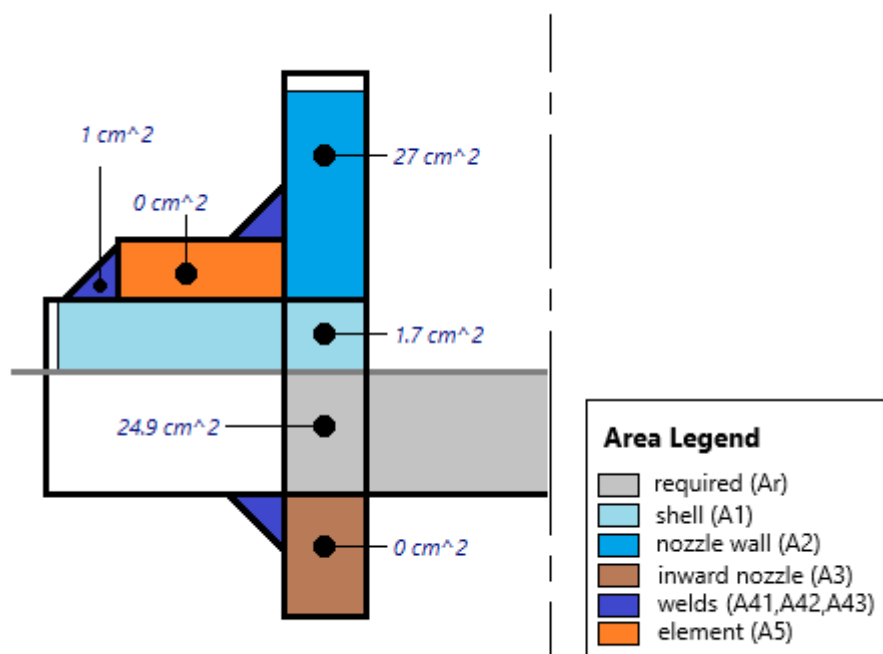


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	24.864	NA	NA
Area in Shell	A1	1.689	NA	NA
Area in Nozzle Wall	A2	27.001	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	29.690	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 43.8 \cdot 1 + 2 \cdot 23.9 \cdot 43.8 \cdot 1 \cdot (1 - 1))$
 $= 24.864 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 138(1 \cdot 45 - 1 \cdot 43.8) - 2 \cdot 23.9$
 $(1 \cdot 45 - 1 \cdot 43.8) \cdot (1 - 1)$
 $= 1.689 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2$
 $= (2 \cdot 59.8)(23.9 - 1.35)1$
 $= 27.001 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 4.3513 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 46.7744 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 46.7744 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 8.2578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.26, \max(46.8, 4.5)]$
 $= 8.2578 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(4.35, 8.26)$
 $= 8.2578 \text{ mm.}$

Available Nozzle Neck Thickness = 26.9240 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:D)

Govrn. thk, $t_g = 26.9$, $tr = 1.35$, $c = 3 \text{ mm.}$, $E* = 1$
 Thickness Ratio $= tr \cdot E* / (t_g - c) = 0.056$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $64.14/102.00 = 0.629$

Weld Size Calculations, Description: K6-2"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (24.9 - 1.69 + 2 \cdot 23.9 \cdot 1 \cdot (1 \cdot 45 - 43.8)) \cdot 138)$
 $= 33436.35 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (27 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 39402.54 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (27 + 0 + 1 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (27 + 0 + 1 + 0 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 105 \cdot 10 \cdot 0.49 \cdot 138$
 $= 11334. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 40.4) \cdot (26.9 - 3) \cdot 0.7 \cdot 138$
 $= 29881. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 105 \cdot (48 - 3) \cdot 0.74 \cdot 138$
 $= 77027. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (11334 + 29881) = 41215 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 0 + 77027 + 0) = 88362 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 77027 + 0) = 88362 \text{ Kgf}$

Summary of Failure Path Calculations:



Path 1-1 = 41215 Kgf, must exceed W = 33436 Kgf or W1 = 39402 Kgf
Path 2-2 = 88361 Kgf, must exceed W = 33436 Kgf or W2 = 69700 Kgf
Path 3-3 = 88361 Kgf, must exceed W = 33436 Kgf or W3 = 69700 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 64 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.5014 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 209.5013 mm.

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Input, Nozzle Desc: K7-2"

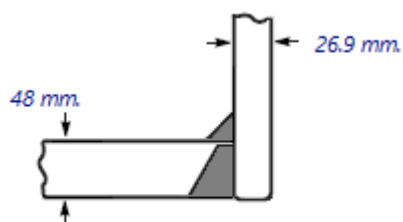
From: 40

Pressure for Reinforcement Calculations	P	64.140	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1825.00	mm.
Shell Finished (Minimum) Thickness	t	48.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		950.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	48.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K7-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	1.060	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 64.1 \cdot 916 / (138 \cdot 1 - 0.6 \cdot 64.1)$
 $= 43.7744$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([64.1 / (138 \cdot 1)] - 1))$
 $= 1.3513$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 194.6480 mm.
 Parallel to Vessel Wall Rn+tn+t 97.3240 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 59.8100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

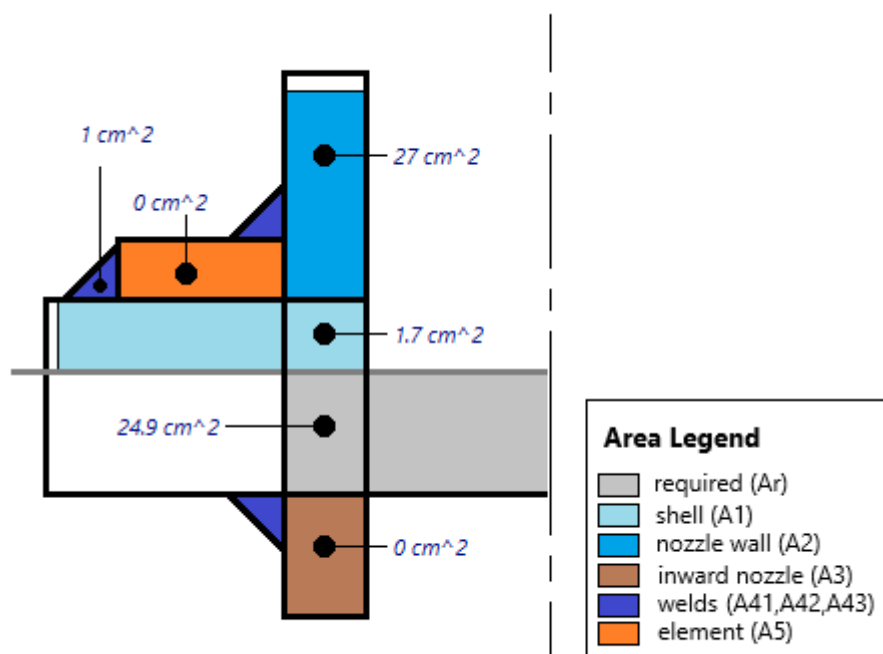


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	24.864	NA	NA
Area in Shell	A1	1.689	NA	NA
Area in Nozzle Wall	A2	27.001	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	29.690	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 43.8 \cdot 1 + 2 \cdot 23.9 \cdot 43.8 \cdot 1 \cdot (1 - 1))$
 $= 24.864 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 138(1 \cdot 45 - 1 \cdot 43.8) - 2 \cdot 23.9$
 $(1 \cdot 45 - 1 \cdot 43.8) \cdot (1 - 1)$
 $= 1.689 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 59.8)(23.9 - 1.35)1$
 $= 27.001 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 4.3513 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 46.7744 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 46.7744 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 8.2578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.26, \max(46.8, 4.5)]$
 $= 8.2578 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(4.35, 8.26)$
 $= 8.2578 \text{ mm.}$

Available Nozzle Neck Thickness = 26.9240 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:D)

Govrn. thk, $t_g = 26.9$, $tr = 1.35$, $c = 3 \text{ mm.}$, $E* = 1$
 Thickness Ratio $= tr \cdot E* / (t_g - c) = 0.056$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $64.14/102.00 = 0.629$

Weld Size Calculations, Description: K7-2"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (24.9 - 1.69 + 2 \cdot 23.9 \cdot 1 \cdot (1 \cdot 45 - 43.8)) \cdot 138)$
 $= 33436.35 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (27 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 39402.54 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (27 + 0 + 1 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (27 + 0 + 1 + 0 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 105 \cdot 10 \cdot 0.49 \cdot 138$
 $= 11334. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 40.4) \cdot (26.9 - 3) \cdot 0.7 \cdot 138$
 $= 29881. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 105 \cdot (48 - 3) \cdot 0.74 \cdot 138$
 $= 77027. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (11334 + 29881) = 41215 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 0 + 77027 + 0) = 88362 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 77027 + 0) = 88362 \text{ Kgf}$

Summary of Failure Path Calculations:



Path 1-1 = 41215 Kgf, must exceed W = 33436 Kgf or W1 = 39402 Kgf
Path 2-2 = 88361 Kgf, must exceed W = 33436 Kgf or W2 = 69700 Kgf
Path 3-3 = 88361 Kgf, must exceed W = 33436 Kgf or W3 = 69700 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 64 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.5014 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 209.5013 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 97.32 mm.

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Input, Nozzle Desc: K8-2"

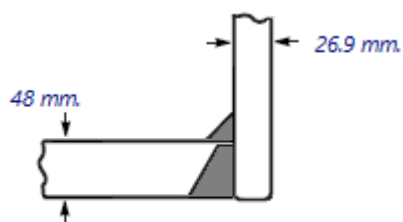
From: 40

Pressure for Reinforcement Calculations	P	64.140	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1825.00	mm.
Shell Finished (Minimum) Thickness	t	48.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		200.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	48.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K8-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	1.060	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 64.1 \cdot 916 / (138 \cdot 1 - 0.6 \cdot 64.1)$
 $= 43.7744$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([64.1 / (138 \cdot 1)] - 1))$
 $= 1.3513$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 194.6480 mm.
 Parallel to Vessel Wall Rn+tn+t 97.3240 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 59.8100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

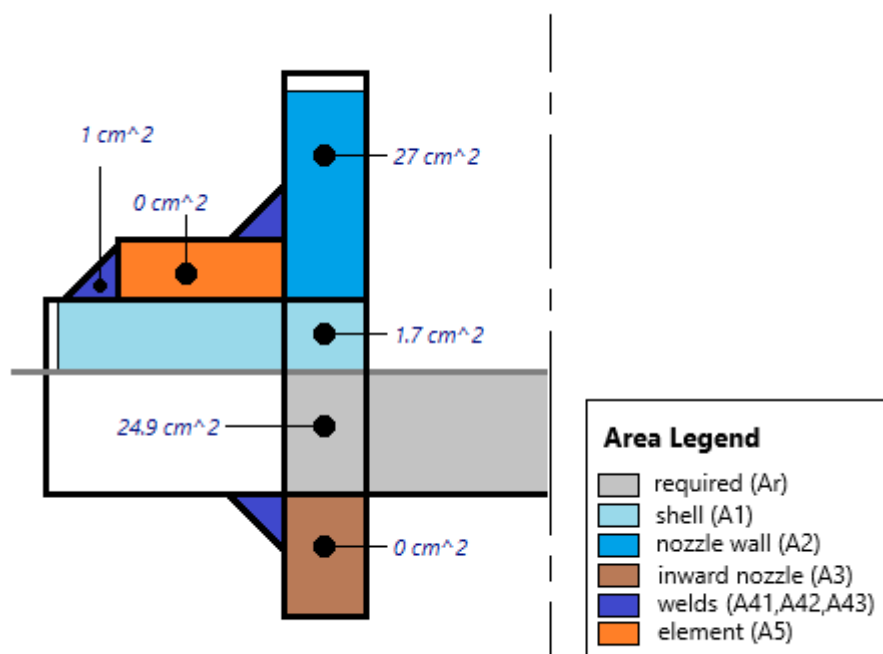


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	24.864	NA	NA
Area in Shell	A1	1.689	NA	NA
Area in Nozzle Wall	A2	27.001	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	29.690	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 43.8 \cdot 1 + 2 \cdot 23.9 \cdot 43.8 \cdot 1 \cdot (1 - 1))$
 $= 24.864 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 138(1 \cdot 45 - 1 \cdot 43.8) - 2 \cdot 23.9$
 $(1 \cdot 45 - 1 \cdot 43.8) \cdot (1 - 1)$
 $= 1.689 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 59.8)(23.9 - 1.35)1$
 $= 27.001 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 4.3513 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 46.7744 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 46.7744 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 8.2578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.26, \max(46.8, 4.5)]$
 $= 8.2578 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(4.35, 8.26)$
 $= 8.2578 \text{ mm.}$

Available Nozzle Neck Thickness = 26.9240 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:D)

Govrn. thk, $tg = 26.9$, $tr = 1.35$, $c = 3 \text{ mm.}$, $E* = 1$
 Thickness Ratio $= tr \cdot E*/(tg - c) = 0.056$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $64.14/102.00 = 0.629$

Weld Size Calculations, Description: K8-2"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (24.9 - 1.69 + 2 \cdot 23.9 \cdot 1 \cdot (1 \cdot 45 - 43.8)) \cdot 138)$
 $= 33436.35 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (27 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 39402.54 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (27 + 0 + 1 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (27 + 0 + 1 + 0 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 105 \cdot 10 \cdot 0.49 \cdot 138$
 $= 11334. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 40.4) \cdot (26.9 - 3) \cdot 0.7 \cdot 138$
 $= 29881. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 105 \cdot (48 - 3) \cdot 0.74 \cdot 138$
 $= 77027. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (11334 + 29881) = 41215 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 0 + 77027 + 0) = 88362 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 77027 + 0) = 88362 \text{ Kgf}$

Summary of Failure Path Calculations:



Path 1-1 = 41215 Kgf, must exceed W = 33436 Kgf or W1 = 39402 Kgf
Path 2-2 = 88361 Kgf, must exceed W = 33436 Kgf or W2 = 69700 Kgf
Path 3-3 = 88361 Kgf, must exceed W = 33436 Kgf or W3 = 69700 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 64 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.5014 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 209.5013 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 97.32 mm.

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Input, Nozzle Desc: K9-2"

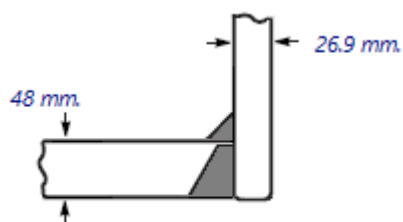
From: 40

Pressure for Reinforcement Calculations	P	64.140	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1825.00	mm.
Shell Finished (Minimum) Thickness	t	48.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		950.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	48.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K9-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	1.060	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 64.1 \cdot 916 / (138 \cdot 1 - 0.6 \cdot 64.1)$
 $= 43.7744$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([64.1 / (138 \cdot 1)] - 1))$
 $= 1.3513$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 194.6480 mm.
 Parallel to Vessel Wall Rn+tn+t 97.3240 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 59.8100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

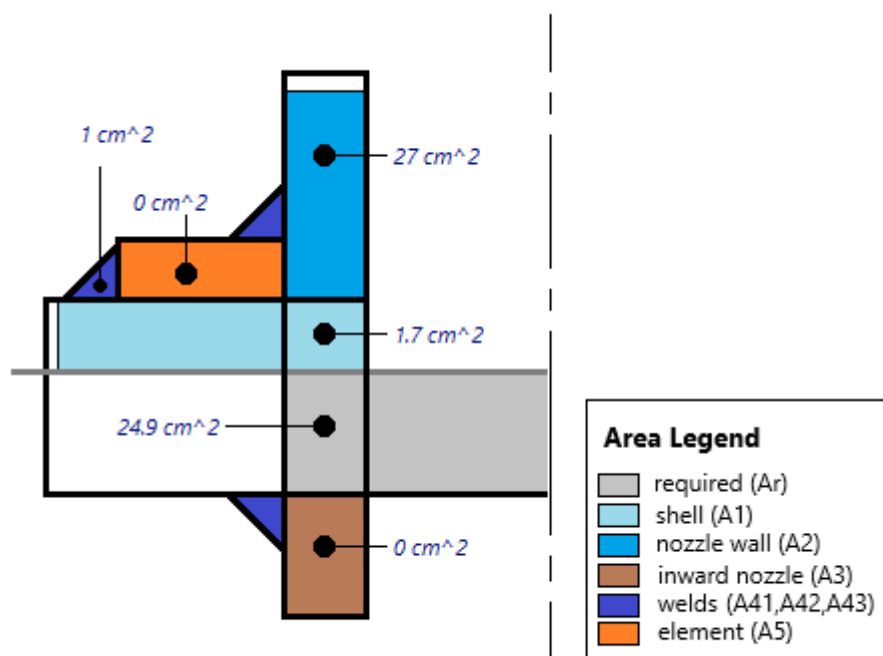


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	24.864	NA	NA
Area in Shell	A1	1.689	NA	NA
Area in Nozzle Wall	A2	27.001	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	29.690	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (56.8 \cdot 43.8 \cdot 1 + 2 \cdot 23.9 \cdot 43.8 \cdot 1 \cdot (1 - 1))$$

$$= 24.864 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 138(1 \cdot 45 - 1 \cdot 43.8) - 2 \cdot 23.9$$

$$(1 \cdot 45 - 1 \cdot 43.8) \cdot (1 - 1)$$

$$= 1.689 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 59.8)(23.9 - 1.35)1$$

$$= 27.001 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.3513 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 46.7744 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 46.7744 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 8.2578 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.26, \max(46.8, 4.5)]$$

$$= 8.2578 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.35, 8.26)$$

$$= 8.2578 \text{ mm.}$$

Available Nozzle Neck Thickness = 26.9240 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:D)

Govrn. thk, tg = 26.9, tr = 1.35, c = 3 mm., E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.056, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $64.14/102.00 = 0.629$

Weld Size Calculations, Description: K9-2"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (24.9 - 1.69 + 2 \cdot 23.9 \cdot 1 \cdot (1 \cdot 45 - 43.8)) \cdot 138)$
 $= 33436.35 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (27 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 39402.54 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (27 + 0 + 1 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (27 + 0 + 1 + 0 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 105 \cdot 10 \cdot 0.49 \cdot 138$
 $= 11334. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 40.4) \cdot (26.9 - 3) \cdot 0.7 \cdot 138$
 $= 29881. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 105 \cdot (48 - 3) \cdot 0.74 \cdot 138$
 $= 77027. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (11334 + 29881) = 41215 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 0 + 77027 + 0) = 88362 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 77027 + 0) = 88362 \text{ Kgf}$

Summary of Failure Path Calculations:



Path 1-1 = 41215 Kgf, must exceed W = 33436 Kgf or W1 = 39402 Kgf
Path 2-2 = 88361 Kgf, must exceed W = 33436 Kgf or W2 = 69700 Kgf
Path 3-3 = 88361 Kgf, must exceed W = 33436 Kgf or W3 = 69700 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 64 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.5014 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 209.5013 mm.

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Input, Nozzle Desc: K10-2"

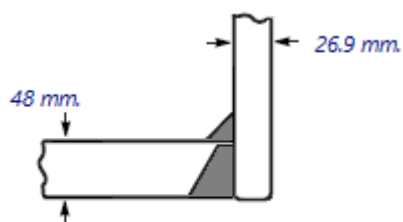
From: 40

Pressure for Reinforcement Calculations	P	64.140	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1825.00	mm.
Shell Finished (Minimum) Thickness	t	48.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		200.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	48.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K10-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	1.060	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 64.1 \cdot 916 / (138 \cdot 1 - 0.6 \cdot 64.1)$
 $= 43.7744$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([64.1 / (138 \cdot 1)] - 1))$
 $= 1.3513$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 194.6480 mm.
 Parallel to Vessel Wall Rn+tn+t 97.3240 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 59.8100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

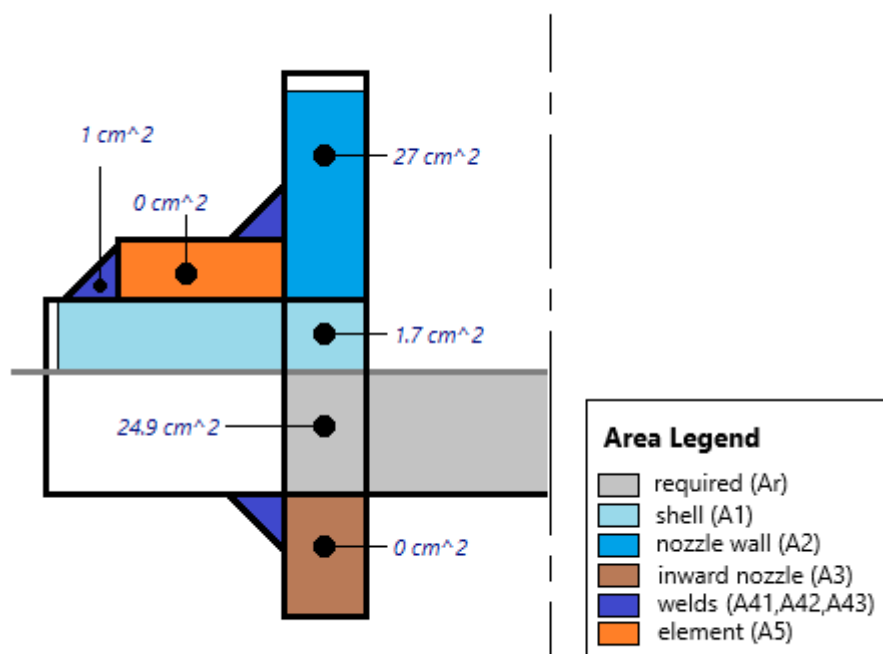


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	24.864	NA	NA
Area in Shell	A1	1.689	NA	NA
Area in Nozzle Wall	A2	27.001	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	29.690	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 43.8 \cdot 1 + 2 \cdot 23.9 \cdot 43.8 \cdot 1 \cdot (1 - 1))$
 $= 24.864 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 138(1 \cdot 45 - 1 \cdot 43.8) - 2 \cdot 23.9$
 $(1 \cdot 45 - 1 \cdot 43.8) \cdot (1 - 1)$
 $= 1.689 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 59.8)(23.9 - 1.35)1$
 $= 27.001 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 4.3513 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 46.7744 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 46.7744 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 8.2578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.26, \max(46.8, 4.5)]$
 $= 8.2578 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(4.35, 8.26)$
 $= 8.2578 \text{ mm.}$

Available Nozzle Neck Thickness = 26.9240 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:D)

Govrn. thk, $t_g = 26.9$, $tr = 1.35$, $c = 3 \text{ mm.}$, $E* = 1$
 Thickness Ratio $= tr \cdot E* / (t_g - c) = 0.056$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $64.14/102.00 = 0.629$

Weld Size Calculations, Description: K10-2"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (24.9 - 1.69 + 2 \cdot 23.9 \cdot 1 \cdot (1 \cdot 45 - 43.8)) \cdot 138)$
 $= 33436.35 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (27 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 39402.54 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (27 + 0 + 1 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (27 + 0 + 1 + 0 + (21.5)) \cdot 138$
 $= 69700.88 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 105 \cdot 10 \cdot 0.49 \cdot 138$
 $= 11334. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 40.4) \cdot (26.9 - 3) \cdot 0.7 \cdot 138$
 $= 29881. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 105 \cdot (48 - 3) \cdot 0.74 \cdot 138$
 $= 77027. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (11334 + 29881) = 41215 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 0 + 77027 + 0) = 88362 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (11334 + 77027 + 0) = 88362 \text{ Kgf}$

Summary of Failure Path Calculations:



Path 1-1 = 41215 Kgf, must exceed W = 33436 Kgf or W1 = 39402 Kgf
Path 2-2 = 88361 Kgf, must exceed W = 33436 Kgf or W2 = 69700 Kgf
Path 3-3 = 88361 Kgf, must exceed W = 33436 Kgf or W3 = 69700 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 64 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.5014 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 209.5013 mm.

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Input, Nozzle Desc: M1-24"

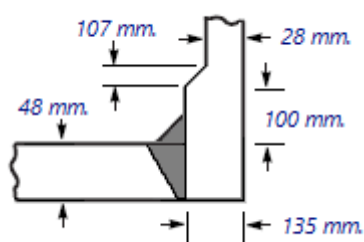
From: 40

Pressure for Reinforcement Calculations	P	64.140	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1825.00	mm.
Shell Finished (Minimum) Thickness	t	48.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1050.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2 +IT	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		225.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	28.0000	mm.
Flange Material		SA-105 +IT	
Hub Height of Integral Nozzle	h	100.0000	mm.
Height of Beveled Transition	L'	107.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		135.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		48.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: M1-24"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 24.000 in.
 Actual Thickness Used in Calculation 1.102 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 64.1 \cdot 916 / (138 \cdot 1 - 0.6 \cdot 64.1)$
 $= 43.7744$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 64.1 \cdot 308 / (138 \cdot 1 - 0.6 \cdot 64.1)$
 $= 14.7174$ mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 107 / 0.5773$
 $= 185.3295$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

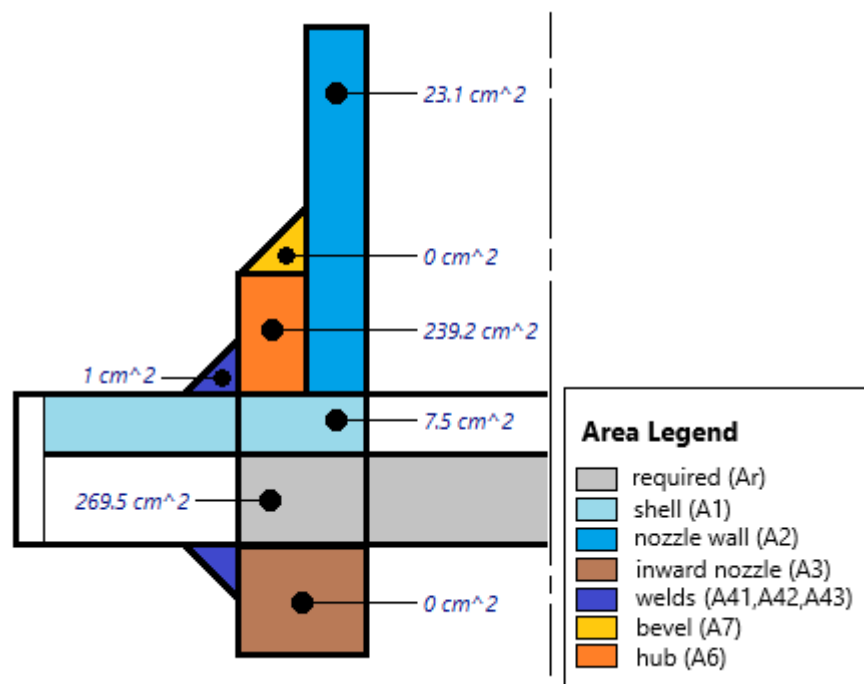
UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1231.2001	mm.
Parallel to Vessel Wall, opening length	d	615.6000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	112.5000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	269.475	NA	NA
Area in Shell	A1	7.545	NA	NA
Area in Nozzle Wall	A2	23.136	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	239.188	NA	NA
TOTAL AREA AVAILABLE	Atot	270.868	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (616 \cdot 43.8 \cdot 1 + 2 \cdot 132 \cdot 43.8 \cdot 1 \cdot (1 - 1))$
 $= 269.475 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 616 (1 \cdot 45 - 1 \cdot 43.8) - 2 \cdot 132$
 $(1 \cdot 45 - 1 \cdot 43.8) \cdot (1 - 1)$
 $= 7.545 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 113) (25 - 14.7) 1$
 $= 23.136 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$



Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$
 $= (2 \cdot \min(113, 350, 100)) \cdot (135 - 28) \cdot 1$
 $= 239.188 \text{ cm}^2$ Includes Bevel Area of 25.188 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 17.7174 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= t_a$
 $= 17.7174 \text{ mm.}$

Available Nozzle Neck Thickness = 28.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: D

Govrn. thk, $t_g = 48$, $t_r = 43.8$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.97$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -22 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -22 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $64.14/102.00 = 0.629$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 107/0.5773$
 $= 185.3295 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 923.4000 mm.
 Parallel to Vessel Wall, opening length d 461.7000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 112.5000 mm.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		179.659	179.659
Area Available in Shell A1		3.772	3.772
Area Available in Nozzle Wall A2		23.136	23.136
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		1.000	1.000
Area Available in Pad A5		0.000	0.000
Area Available in Hub A6		239.188	239.188
TOTAL AREA AVAILABLE Atot		267.096	267.096

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (269 - 7.54 + 2 \cdot 132 \cdot 1 \cdot (1 \cdot 45 - 43.8)) \cdot 138)$
 $= 373137.03 \text{ Kgf}$

For hub type nozzles, A₂ includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_4 - (W_i - \text{Can}) / 0.707)^2 \cdot f_{r2} \cdot S_v$
 $= (262 + 1 - 0 \cdot 1) \cdot 138$
 $= 370544.22 \text{ Kgf}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (262 + 0 + 1 + (119)) \cdot 138$
 $= 537717.50 \text{ Kgf}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (262 + 0 + 1 + 0 + (119)) \cdot 138$
 $= 537717.50 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 880 \cdot 10 \cdot 0.49 \cdot 138$
 $= 95269. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 374) \cdot (135 - 3) \cdot 0.7 \cdot 138$
 $= 1526903. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 880 \cdot (48 - 3) \cdot 0.74 \cdot 138$
 $= 647440. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (95269 + 1526903) = 1622172 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (95269 + 0 + 647440 + 0) = 742709 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (95269 + 647440 + 0) = 742709 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 1622172 Kgf, must exceed W = 373137 Kgf	or W1 = 370544 Kgf
Path 2-2 = 742709 Kgf, must exceed W = 373137 Kgf	or W2 = 537717 Kgf
Path 3-3 = 742709 Kgf, must exceed W = 373137 Kgf	or W3 = 537717 Kgf



Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 64 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 112.9800 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 510.9799 mm.

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Input, Nozzle Desc: N2-16"

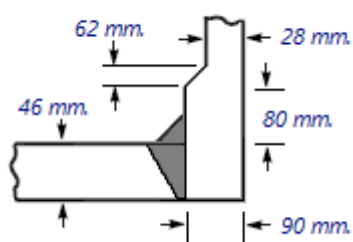
From: 50

Pressure for Reinforcement Calculations	P	64.140	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	1825.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	46.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2 +IT	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	28.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	80.0000	mm.
Height of Beveled Transition	L'	62.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	46.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: N2-16"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 16.000 in.
Actual Thickness Used in Calculation 1.102 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $64.1 \cdot 1831 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 64.1)$
= 38.3570 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $64.1 \cdot 206 / (138 \cdot 1 - 0.6 \cdot 64.1)$
= 9.8596 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 62/0.5773
= 107.3872 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

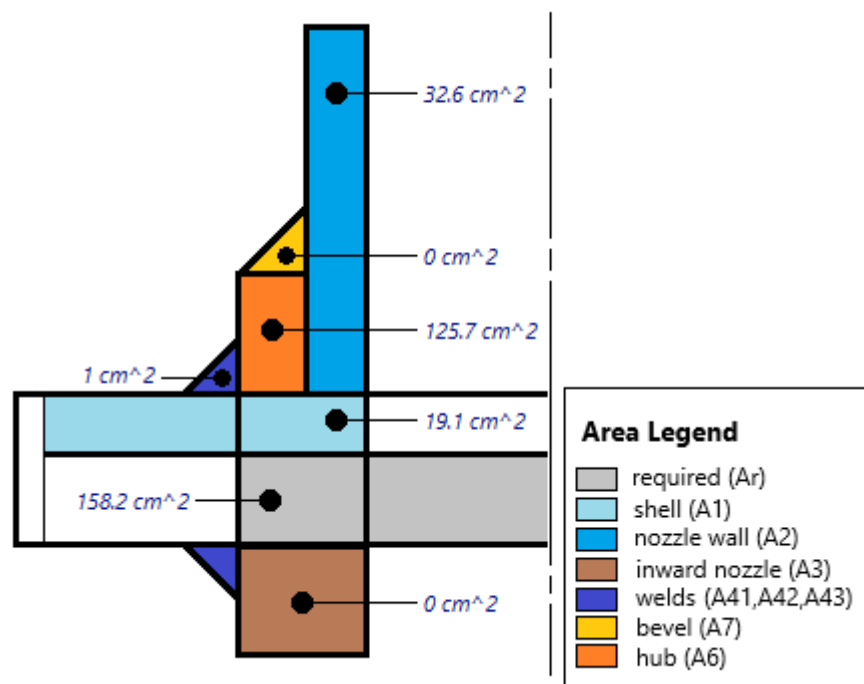
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	824.8000	mm.
Parallel to Vessel Wall, opening length	d	412.4000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	107.5000	mm.

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 138/138)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 138/138)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1, 1)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	158.186	NA	NA
Area in Shell	A1	19.147	NA	NA
Area in Nozzle Wall	A2	32.551	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	125.735	NA	NA
TOTAL AREA AVAILABLE	Atot	178.433	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (412 \cdot 38.4 \cdot 1 + 2 \cdot 87 \cdot 38.4 \cdot 1 \cdot (1 - 1))$$

$$= 158.186 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 412 (1 \cdot 43 - 1 \cdot 38.4) - 2 \cdot 87 (1 \cdot 43 - 1 \cdot 38.4) \cdot (1 - 1)$$

$$= 19.147 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 107) (25 - 9.86) \cdot 1$$

$$= 32.551 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$



Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$
 $= (2 \cdot \min(107, 500, 80)) \cdot (90 - 28) \cdot 1$
 $= 125.735 \text{ cm}^2$ Includes Bevel Area of 26.537 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 12.8596 \text{ mm.}$
 Wall Thickness per UG16(b), $tr_{16b} = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb_1 = 45.5664 \text{ mm.}$
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 45.5664 \text{ mm.}$
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb_3 = 11.3312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[11.3, \max(45.6, 4.5)]$
 $= 11.3312 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(12.9, 11.3)$
 $= 12.8596 \text{ mm.}$

Available Nozzle Neck Thickness = 28.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	38.7,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	306.3 N./mm ²	Passed
Occasional	:	24.9,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	7.8,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $tg = 28$, $tr = 9.86$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio $= tr \cdot E^* / (tg - c) = 0.39$, Temp. Reduction = 52 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-30 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govrn. thk, $tg = 46$, $tr = 38.4$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio $= tr \cdot E^* / (tg - c) = 0.89$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -27 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $64.14 / 102.00 = 0.629$

Weld Size Calculations, Description: N2-16"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm.
---	------	-------------

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (158 - 19.1 + 2 \cdot 87 \cdot 1 \cdot (1 \cdot 43 - 38.4)) \cdot 138)$
 $= 207020.59 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (158 + 1 - 0 \cdot 1) \cdot 138$
 $= 224144.22 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (158 + 0 + 1 + (74.8)) \cdot 138$
 $= 329427.31 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (158 + 0 + 1 + 0 + (74.8)) \cdot 138$
 $= 329427.31 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 586 \cdot 10 \cdot 0.49 \cdot 138$
 $= 63511. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 250) \cdot (90 - 3) \cdot 0.7 \cdot 138$
 $= 672244. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 586 \cdot (46 - 3) \cdot 0.74 \cdot 138$
 $= 412443. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (63511 + 672244) = 735755 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (63511 + 0 + 412443 + 0) = 475955 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (63511 + 412443 + 0) = 475955 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 735754 Kgf, must exceed W = 207020 Kgf or W1 = 224144 Kgf
 Path 2-2 = 475954 Kgf, must exceed W = 207020 Kgf or W2 = 329427 Kgf
 Path 3-3 = 475954 Kgf, must exceed W = 207020 Kgf or W3 = 329427 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 64 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.



The Drop for this Nozzle is : 25.7940 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 571.7939 mm.

Input Echo, WRC107/537 Item 1, Description: N2-16" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	3285.000	mm.
Vessel Thickness	Tv	46.000	mm.
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	406.400	mm.
Nozzle Thickness	Tn	90.000	mm.
Nozzle Material		SA-266 2	
Nozzle UNS Number		K03506	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²
Design Internal Pressure	Dp	64.140	bars
Include Pressure Thrust		No	

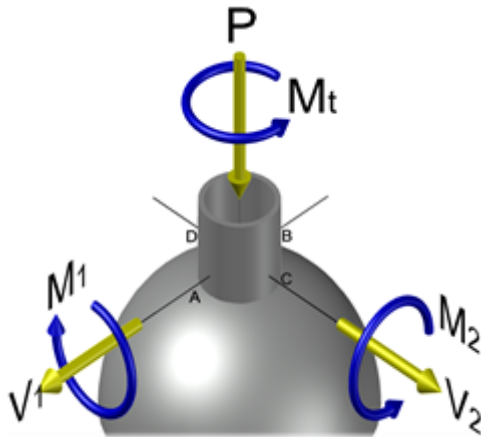
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	3669.8	Kgf
Longitudinal Shear (SUS)	(V1) V1	2650.4	Kgf
Circumferential Shear (SUS)	(Vc) V2	2650.4	Kgf
Circumferential Moment (SUS)	(Mc) M1	3262.2	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	3262.2	Kg-m.
Torsional Moment (SUS)	Mt	3751.5	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 586.4 + 2 \cdot 1.65 \cdot \sqrt{(1667.0 (46.0 - 3.0))}$$

$$= 1469.919 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	3669.8	Kgf
Circumferential Shear	(VC) V2	2650.4	Kgf
Longitudinal Shear	(VL) V1	2650.4	Kgf
Circumferential Moment	(MC) M1	3262.2	Kg-m.
Longitudinal Moment	(ML) M2	3262.2	Kg-m.
Torsional Moment	MT	3751.5	Kg-m.

Dimensionless Param: U = 1.10 TAU = 5.00 (2.87) RHO = 0.49

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.03692	(A,B,C,D)
$M(x) / P$	SP 1	0.03807	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.04268	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.06569	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.04268	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.06569	(A,B,C,D)
$N(y) * T / P$	SP 1	0.05978	(A,B,C,D)
$M(y) / P$	SP 1	0.01106	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.05898	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.02121	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.05898	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.02121	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7
Rad. Bend.	P	-4.4	4.4	-4.4	4.4	-4.4	4.4	-4.4	4.4
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-2.8	-2.8	2.8	2.8
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-25.5	25.5	25.5	-25.5
Rad. Memb.	ML	-2.8	-2.8	2.8	2.8	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-25.5	25.5	25.5	-25.5	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-33.4	26.4	23.1	-19.0	-33.4	26.4	23.1	-19.0



Tang. Memb. P	-1.2	-1.2	-1.2	-1.2	-1.2	-1.2	-1.2	-1.2
Tang. Bend. P	-1.3	1.3	-1.3	1.3	-1.3	1.3	-1.3	1.3
Tang. Memb. MC	0.0	0.0	0.0	0.0	-3.8	-3.8	3.8	3.8
Tang. Bend. MC	0.0	0.0	0.0	0.0	-8.2	8.2	8.2	-8.2
Tang. Memb. ML	-3.8	-3.8	3.8	3.8	0.0	0.0	0.0	0.0
Tang. Bend. ML	-8.2	8.2	8.2	-8.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-14.5	4.5	9.6	-4.3	-14.5	4.5	9.6	-4.3
Shear VC	0.7	0.7	-0.7	-0.7	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Shear MT	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6
Tot. Shear	2.2	2.2	0.9	0.9	0.9	0.9	2.2	2.2
Str. Int.	33.7	26.7	23.1	19.0	33.4	26.5	23.4	19.3

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		122.8	122.8	122.8	122.8	122.8	122.8	122.8	122.8
Rad. Pl (SUS)		-3.5	-3.5	2.0	2.0	-3.5	-3.5	2.0	2.0
Rad. Q (SUS)		-29.9	29.9	21.0	-21.0	-29.9	29.9	21.0	-21.0
Long. Pm (SUS)		122.8	122.8	122.8	122.8	122.8	122.8	122.8	122.8
Long. Pl (SUS)		-5.0	-5.0	2.6	2.6	-5.0	-5.0	2.6	2.6
Long. Q (SUS)		-9.5	9.5	6.9	-6.9	-9.5	9.5	6.9	-6.9
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.7	0.7	-0.7	-0.7	-0.7	-0.7	0.7	0.7
Shear Q (SUS)		1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6
Pm (SUS)		122.8	122.8	122.8	122.8	122.8	122.8	122.8	122.8
Pm+Pl (SUS)		119.6	119.6	125.9	125.9	119.6	119.6	125.9	125.9
Pm+Pl+Q (Total)		108.6	149.5	145.9	118.6	108.3	149.3	146.2	118.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	122.79	138.00	Passed
Pm+Pl (SUS)	125.85	207.00	Passed
Pm+Pl+Q (TOTAL)	149.46	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K11-2"

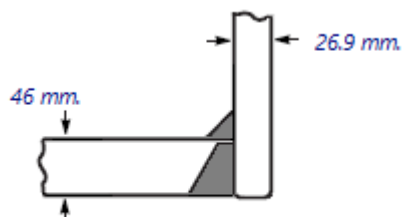
From: 50

Pressure for Reinforcement Calculations	P	64.140	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	1825.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	46.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	650.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	650.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	46.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K11-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 1.060 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 64.1 \cdot 1831 \cdot 1 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 64.1)$
 $= 42.5664$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([64.1 / (138 \cdot 1)] - 1))$
 $= 1.3513$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 197.4945 mm.
 Parallel to Vessel Wall Rn+tn+t 98.7473 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 59.8100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

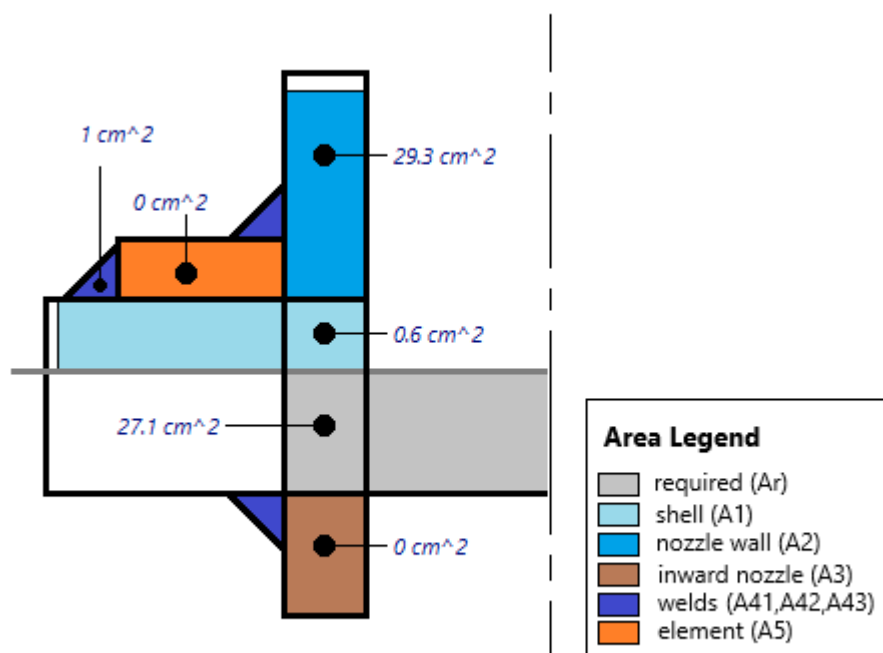


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	27.092	NA	NA



Area in Shell	A1	0.580	NA	NA
Area in Nozzle Wall	A2	29.349	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	30.929	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 63.18 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (63.6 \cdot 42.6 \cdot 1 + 2 \cdot 23.9 \cdot 42.6 \cdot 1 \cdot (1 - 1))$
 $= 27.092 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 134(1 \cdot 43 - 1 \cdot 42.6) - 2 \cdot 23.9$
 $(1 \cdot 43 - 1 \cdot 42.6) \cdot (1 - 1)$
 $= 0.580 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 59.8)(23.9 - 1.35)/\sin(66.9)$
 $= 29.349 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:D)

Govrn. thk, tg = 26.9, tr = 1.35, c = 3 mm., E* = 1
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.056$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $64.14/102.00 = 0.629$

Weld Size Calculations, Description: K11-2"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 \cdot Wo$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f r_1 \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (27.1 - 0.58 + 2 \cdot 23.9 \cdot 1 \cdot \\
 &\quad (1 \cdot 43 - 42.6)) \cdot 138) \\
 &= 37599.29 \text{ Kgf}
 \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f r_2) \cdot S_v \\
 &= (29.3 + 0 + 1 - 0 \cdot 1) \cdot 138 \\
 &= 42706.18 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S_v \\
 &= (29.3 + 0 + 1 + (20.6)) \cdot 138 \\
 &= 71657.93 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S \\
 &= (29.3 + 0 + 1 + 0 + (20.6)) \cdot 138 \\
 &= 71657.93 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 117 \cdot 10 \cdot 0.49 \cdot 138 \\
 &= 12700. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 45.2) \cdot (26.9 - 3) \cdot 0.7 \cdot 138 \\
 &= 33483. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 117 \cdot (46 - 3) \cdot 0.74 \cdot 138 \\
 &= 82476. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (12700 + 33483) = 46183 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (12700 + 0 + 82476 + 0) = 95176 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\
 &= (12700 + 82476 + 0) = 95176 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 46183	Kgf, must exceed W = 37599 Kgf	or W1 = 42706 Kgf
Path 2-2 = 95176	Kgf, must exceed W = 37599 Kgf	or W2 = 71657 Kgf
Path 3-3 = 95176	Kgf, must exceed W = 37599 Kgf	or W3 = 71657 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 64 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K11-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 1.060 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 64.1 \cdot 1831 \cdot 1 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 64.1)$
 $= 42.5664 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 (\exp([64.1 / (138 \cdot 1)] - 1))$
 $= 1.3513 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 190.6480 mm.
 Parallel to Vessel Wall Rn+tn+t 95.3240 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 59.8100 mm.

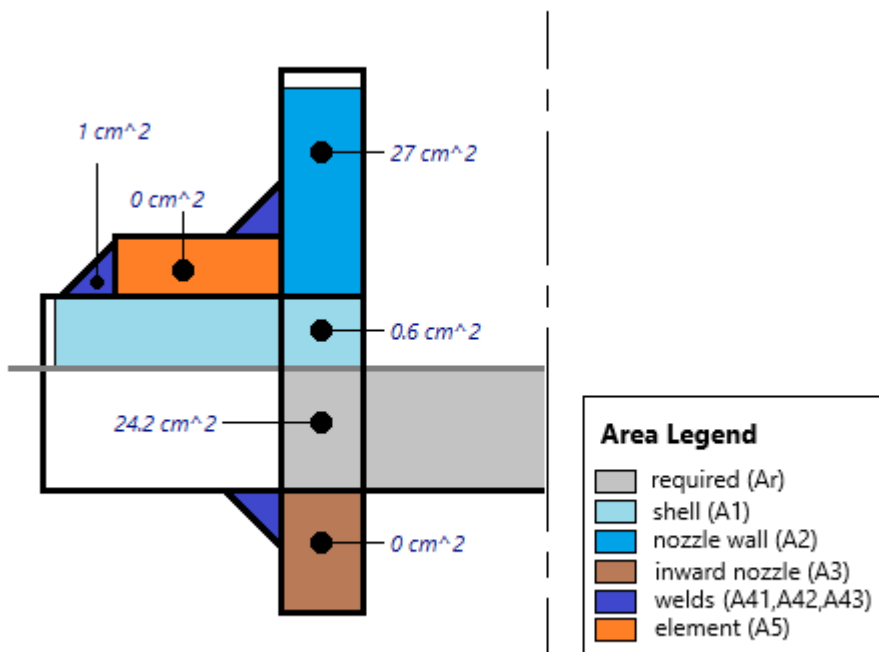


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	24.178	NA	NA
Area in Shell	A1	0.580	NA	NA
Area in Nozzle Wall	A2	27.001	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	28.581	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)

$$= (56.8 \cdot 42.6 \cdot 1 + 2 \cdot 23.9 \cdot 42.6 \cdot 1 \cdot (1-1))$$

$$= 24.178 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 134 (1 \cdot 43 - 1 \cdot 42.6) - 2 \cdot 23.9$$

$$(1 \cdot 43 - 1 \cdot 42.6) \cdot (1 - 1)$$

$$= 0.580 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 59.8) (23.9 - 1.35) 1$$

$$= 27.001 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi \cdot can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.3513 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 45.5664 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 45.5664 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 8.2578 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.26, \max(45.6, 4.5)]$$

$$= 8.2578 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.35, 8.26)$$

$$= 8.2578 \text{ mm.}$$

Available Nozzle Neck Thickness = 26.9240 mm. -> OK

Weld Size Calculations, Description: K11-2"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (24.2 - 0.58 + 2 \cdot 23.9 \cdot 1 \cdot$$

$$(1 \cdot 43 - 42.6)) 138)$$

$$= 33498.31 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (Wi \cdot Can / 0.707)^2 \cdot fr2) \cdot Sv$$

$$= (27 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 39402.54 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (27 + 0 + 1 + (20.6)) \cdot 138$$

$$= 68354.29 \text{ Kg}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r 1)) \cdot S \\
 &= (27 + 0 + 1 + 0 + (20.6)) \cdot 138 \\
 &= 68354.29 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 105 \cdot 10 \cdot 0.49 \cdot 138 \\
 &= 11334. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 40.4) \cdot (26.9 - 3) \cdot 0.7 \cdot 138 \\
 &= 29881. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 105 \cdot (46 - 3) \cdot 0.74 \cdot 138 \\
 &= 73604. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (11334 + 29881) = 41215 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (11334 + 0 + 73604 + 0) = 84938 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (11334 + 73604 + 0) = 84938 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 41215	Kgf, must exceed W = 33498 Kgf	or W1 = 39402 Kgf
Path 2-2 = 84938	Kgf, must exceed W = 33498 Kgf	or W2 = 68354 Kgf
Path 3-3 = 84938	Kgf, must exceed W = 33498 Kgf	or W3 = 68354 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 64 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 23.0201 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 722.7921 mm.

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Input, Nozzle Desc: K12-2"

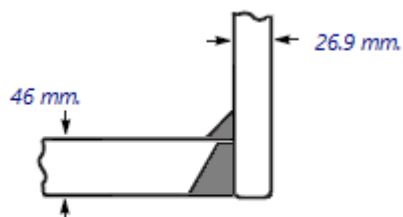
From: 50

Pressure for Reinforcement Calculations	P	64.140	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	1825.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	46.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	650.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	650.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	46.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K12-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 1.060 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 64.1 \cdot 1831 \cdot 1 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 64.1)$
 $= 42.5664$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([64.1 / (138 \cdot 1)] - 1))$
 $= 1.3513$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 197.4945 mm.
 Parallel to Vessel Wall Rn+tn+t 98.7473 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 59.8100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

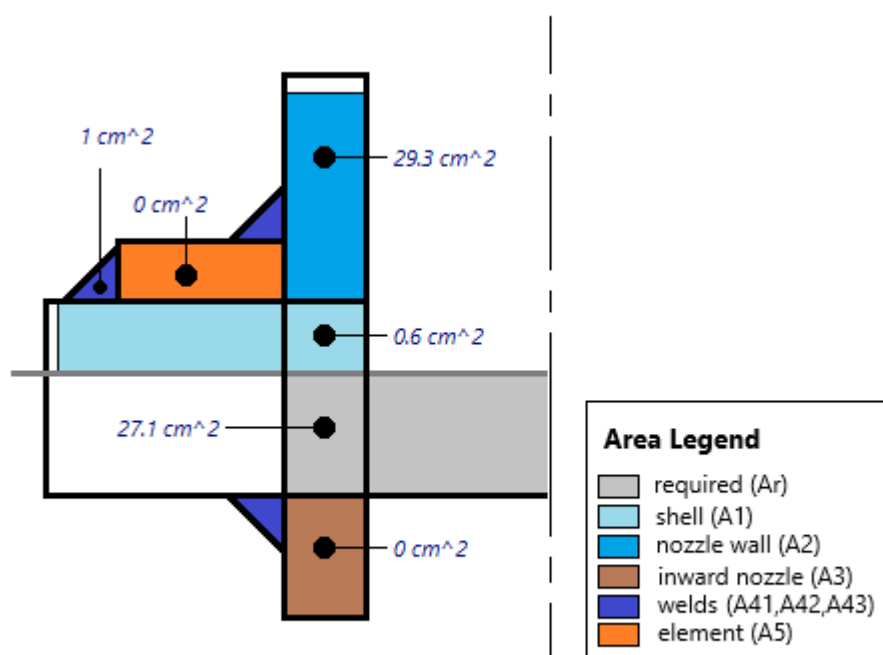


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	27.092	NA	NA



Area in Shell	A1	0.580	NA	NA
Area in Nozzle Wall	A2	29.349	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	30.929	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 63.18 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (63.6 \cdot 42.6 \cdot 1 + 2 \cdot 23.9 \cdot 42.6 \cdot 1 \cdot (1 - 1))$$

$$= 27.092 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 134(1 \cdot 43 - 1 \cdot 42.6) - 2 \cdot 23.9$$

$$(1 \cdot 43 - 1 \cdot 42.6) \cdot (1 - 1)$$

$$= 0.580 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2/\sin(\alpha3)$$

$$= (2 \cdot 59.8)(23.9 - 1.35)/\sin(66.9)$$

$$= 29.349 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:D)

Govrn. thk, tg = 26.9, tr = 1.35, c = 3 mm., E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.056, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 21 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 64.14/102.00 = 0.629$$

Weld Size Calculations, Description: K12-2"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f r_1 \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (27.1 - 0.58 + 2 \cdot 23.9 \cdot 1 \cdot \\
 &\quad (1 \cdot 43 - 42.6)) \cdot 138) \\
 &= 37599.29 \text{ Kgf}
 \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f r_2) \cdot S_v \\
 &= (29.3 + 0 + 1 - 0 \cdot 1) \cdot 138 \\
 &= 42706.18 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S_v \\
 &= (29.3 + 0 + 1 + (20.6)) \cdot 138 \\
 &= 71657.93 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S \\
 &= (29.3 + 0 + 1 + 0 + (20.6)) \cdot 138 \\
 &= 71657.93 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 117 \cdot 10 \cdot 0.49 \cdot 138 \\
 &= 12700. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 45.2) \cdot (26.9 - 3) \cdot 0.7 \cdot 138 \\
 &= 33483. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 117 \cdot (46 - 3) \cdot 0.74 \cdot 138 \\
 &= 82476. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (12700 + 33483) = 46183 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (12700 + 0 + 82476 + 0) = 95176 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\
 &= (12700 + 82476 + 0) = 95176 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 46183	Kgf, must exceed W = 37599 Kgf	or W1 = 42706 Kgf
Path 2-2 = 95176	Kgf, must exceed W = 37599 Kgf	or W2 = 71657 Kgf
Path 3-3 = 95176	Kgf, must exceed W = 37599 Kgf	or W3 = 71657 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 64 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K12-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 1.060 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 64.1 \cdot 1831 \cdot 1 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 64.1)$
 $= 42.5664 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 (\exp([64.1 / (138 \cdot 1)] - 1))$
 $= 1.3513 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 190.6480 mm.
 Parallel to Vessel Wall Rn+tn+t 95.3240 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 59.8100 mm.

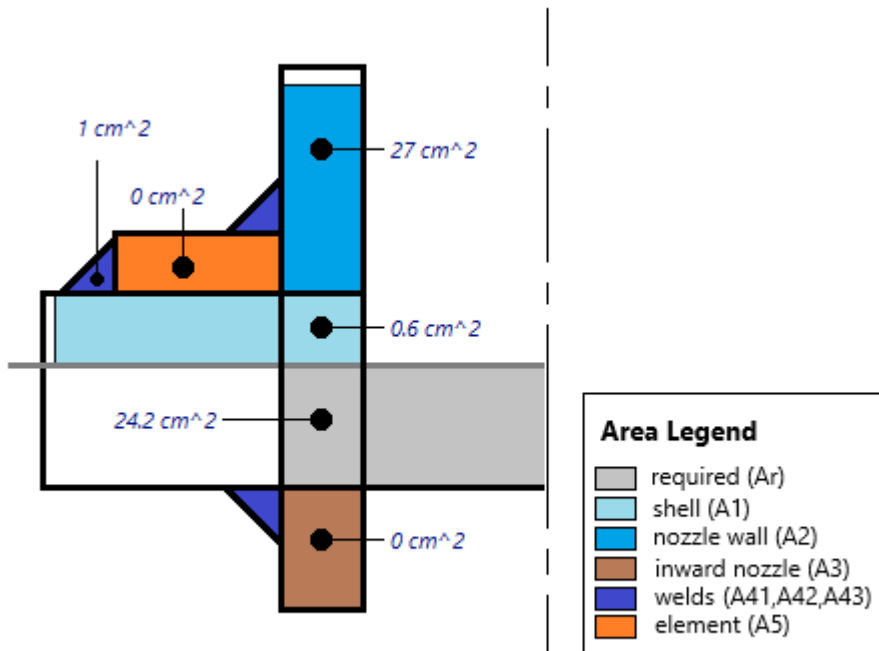


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	24.178	NA	NA
Area in Shell	A1	0.580	NA	NA
Area in Nozzle Wall	A2	27.001	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	28.581	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)



$$= (56.8 \cdot 42.6 \cdot 1 + 2 \cdot 23.9 \cdot 42.6 \cdot 1 \cdot (1-1))$$

$$= 24.178 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 134 (1 \cdot 43 - 1 \cdot 42.6) - 2 \cdot 23.9$$

$$(1 \cdot 43 - 1 \cdot 42.6) \cdot (1 - 1)$$

$$= 0.580 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 59.8) (23.9 - 1.35) \cdot 1$$

$$= 27.001 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi \cdot can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.3513 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 45.5664 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 45.5664 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 8.2578 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.26, \max(45.6, 4.5)]$$

$$= 8.2578 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.35, 8.26)$$

$$= 8.2578 \text{ mm.}$$

Available Nozzle Neck Thickness = 26.9240 mm. -> OK

Weld Size Calculations, Description: K12-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (24.2 - 0.58 + 2 \cdot 23.9 \cdot 1 \cdot$$

$$(1 \cdot 43 - 42.6)) \cdot 138)$$

$$= 33498.31 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (Wi \cdot Can / 0.707)^2 \cdot fr2) \cdot Sv$$

$$= (27 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 39402.54 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (27 + 0 + 1 + (20.6)) \cdot 138$$

$$= 68354.29 \text{ Kg}$$

Weld Load [W3]:



$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r 1)) \cdot S \\
 &= (27 + 0 + 1 + 0 + (20.6)) \cdot 138 \\
 &= 68354.29 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 105 \cdot 10 \cdot 0.49 \cdot 138 \\
 &= 11334. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 40.4) \cdot (26.9 - 3) \cdot 0.7 \cdot 138 \\
 &= 29881. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 105 \cdot (46 - 3) \cdot 0.74 \cdot 138 \\
 &= 73604. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (11334 + 29881) = 41215 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (11334 + 0 + 73604 + 0) = 84938 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (11334 + 73604 + 0) = 84938 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 41215	Kgf, must exceed W = 33498 Kgf	or W1 = 39402 Kgf
Path 2-2 = 84938	Kgf, must exceed W = 33498 Kgf	or W2 = 68354 Kgf
Path 3-3 = 84938	Kgf, must exceed W = 33498 Kgf	or W3 = 68354 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 64 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 23.0201 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 722.7921 mm.

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: